Capital Requirements, Earnings Stability, and Asset Maturity in Banking*

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Abstract

I demonstrate that capital regulation can incentivize banks to maintain stable earnings, as doing so ensures they have sufficient capital to capture profitable lending opportunities. Because deposit spreads rise and long-term asset spreads fall when risk-free rates increase, banks adjust asset maturities to stabilize earnings. When rates are low, the interest rate sensitivity of deposit spreads is higher, implying longer optimal asset maturities. Empirically, I find that this interest rate sensitivity increases more for banks with greater deposit market power, and these banks extended their asset maturities more following the rate reductions induced by the COVID-19 pandemic.

1 Introduction

Banks take interest rate risk management very seriously — they discuss their interest rate risk exposure and estimate the impact of rate shocks on their earnings and capital in every annual report. These risk management practices have historically resulted in remarkably stable earnings, with little correlation to interest rate fluctuations.[1] However, despite these risk management strategies, banks experienced significant losses from their interest rate risk exposure in 2022–2023, which were a key contributor to the 2023 banking turmoil. Moreover, as shown in Figure 1, banks extended asset maturities significantly during the low-interest-rate period preceding the turmoil, with the sector's median asset maturities rising from 3 years in 2007 to nearly 6 years by 2021.[2] These observations raise important questions: Why do banks' risk management strategies prioritize stable earnings? Does this objective lead them to extend asset maturities in low-interest-rate environments?

To answer these questions, this paper develops and tests a stylized banking model that demonstrates how capital regulation incentivizes banks to maintain stable earnings and why long-term assets serve as important tools for banks to maintain stable earnings. Within the model, the incentive to maintain stable earnings explains why maturity mismatches exist, why they differ across banks, and why they vary over time.

The model incorporates three key elements. First, banks' lending businesses exhibit decreasing returns to scale. Second, banks operate under regulatory capital constraints that determine the amount that they can lend. Third, banks possess market power in the deposit market,

[1] A large body of empirical research has documented this pattern across various sample periods and countries. See, for example, Flannery (1981); Flannery (1983); English (2002); Drechsler, Savov, and Schnabl (2021); Begenau and Stafford (2022). In the literature, this phenomenon is also referred to as cash flow hedging or stable cash flows. For clarity and consistency with banks' annual reports, I use the term earnings throughout this paper to represent banks' profitability or overall cash flows.

[2] Notably, the increase in asset maturities happened in low-interest-rate environment and the asset maturities also dropped when risk-free rate increases. The increase was not uniform across banks. For instance, after the Fed funds rate cut induced by the COVID-19 pandemic, the asset maturity of Silicon Valley Bank (SVB) rose sharply from 4.59 years in 2020Q1 to 8.27 years in 2021Q4, while KeyBank's asset maturity increased more modestly from 3.70 to 4.65 years over the same period.

which results in deposit betas, i.e., the pass-through rate from the risk-free rate to deposit rates, being less than one. Moreover, deposit betas are lower when risk-free rate is low, due to the existence of Zero Lower Bound (ZLB).

The first and second elements provide banks with an incentive to maintain stable earnings. If they fail to do so, fluctuations in earnings will create fluctuations in capital, which will result in fluctuations in lending. This outcome is undesirable for value-maximizing banks because lending has decreasing returns to scale.[3] However, because deposit betas are less than one, the spread banks earn on deposits increases with the risk-free rate, introducing fluctuations into banks' earnings.[4] Since the spread between fixed long-term rates and the risk-free rate decreases as the risk-free rate increases, holding long-term assets helps banks stabilize their earnings.

Because deposit betas are lower in low-interest-rate environments, the deposit spread increases faster with the risk-free rate, which incentivizes banks to choose longer asset maturity. Similarly, for a given level of the risk-free rate, banks with greater deposit market power tend to have lower deposit betas, and thus also prefer longer asset maturities. Finally, I show that when the economy enters low-interest-rate environments, deposit betas tend to decline more for banks with greater deposit market power, leading these banks to increase their asset maturity more.[5]

I then empirically measure banks' deposit betas across different interest rate environments. Because real-world banks also have non-deposit liabilities, I measure banks' deposit betas by estimating the interest rate sensitivity of banks' cost of funds, which are defined as interest expense divided by liabilities. This approach follows Drechsler, Savov, and Schnabl (2021) and mitigates the concern that banks could use non-deposit liabilities to maintain stable earnings.

[3]The intuition is similar to Froot, Scharfstein, and Stein (1993), where firms with financial constraint have an incentive to stabilize earnings when investment returns are diminishing. Mathematically, this follows from Jensen's inequality.

[4]As discussed in Section 2, earnings derived from banks' deposits—hereafter referred to as deposit profit—are equal to the product of the deposit spread and deposit quantity. Because deposit quantity varies little across consecutive periods, the primary driver of this component of earnings is the deposit spread.

[5]This last point can be illustrated by comparing a bank with no deposit market power to one with some deposit market power. For the bank with no market power, the deposit beta remains fixed at one. In contrast, the bank with deposit market power has a beta below one, which declines further when the risk-free rate is low.

The estimation results indicate that the average deposit beta is 0.370 when the Fed funds rate is relatively high and 0.229 in low-interest-rate environments. However, banks' earnings, as measured by return on assets, do not vary with the risk-free rate in either high or low-interest-rate environments. Together, these findings provide initial evidence that banks extend their asset maturities in low-interest-rate environments to stabilize earnings.

An important implication of the model is that banks' deposit market power determines how their asset maturities react to changes in the Fed funds rates. Specifically, banks with greater deposit market power will extend their asset maturity more when the economy enters low-interest-rate environments. To test this implication, I first classify banks into high and low deposit market power groups based on their average deposit rate pass-through during a given rate-hiking cycle. To do so, I employ a mixture model approach, which is commonly used in the machine learning literature, to jointly estimate both the average pass-through rates and the posterior probability of each bank's membership in a particular group. Group membership is then assigned based on the posterior probability. [6]

I apply this method using data from the 2015–2018 period and find that it effectively separates the banks into two groups, with the posterior probability of being high deposit market power banks concentrated around 0 and 1. To ensure that the classification captures a characteristic that remains relatively stable over time, I apply the method to the 2022–2024 period, when the recent monetary policy tightening occurred, and find that the classification results are consistent across both periods.

With the classification results, I estimate deposit betas for two groups of banks across environments with relatively high and low risk-free rates. I find that the average deposit beta for banks with low deposit market power is 0.454 when the risk-free rate is high and decreases to 0.337 when the economy enters low-interest-rate environments, a decrease of 0.117. In contrast, the average deposit beta for banks with high deposit market power is 0.325 when the risk-free

[6] Specifically, I designate the group with the lower pass-through rate as high deposit market power banks. Banks with a posterior probability above 0.8 of belonging to this group are classified as high deposit market power banks, while those with a probability below 0.2 are classified as low deposit market power banks.

rate is high and decreases to 0.152 when the economy enters low-interest-rate environments, a decrease of 0.173.

The differential change in deposit betas implies that banks with greater deposit market power are likely to extend their asset maturity more. To test this implication, I perform a Difference-in-Differences analysis that estimates changes in the asset maturity of banks with high versus low deposit market power, before and after the Fed funds rate cut induced by the COVID-19 pandemic in 2020. Confirming the model prediction, I find that while both groups of banks extended their asset maturity following the interest rate shock, banks with high deposit market power increased their asset maturity by 32.2% more than those with low deposit market power.

One implicit assumption of the model is that profits from lending do not introduce interest rate risk to banks' earnings. While this assumption may not hold empirically, due to factors such as fluctuations in credit demand and changes in credit risk, the main results remain valid as long as this risk does not vary significantly when the economy enters a low-interest-rate environment. To validate this assumption, I estimate the interest rate sensitivity of banks' interest income from loans, net of charge-offs, for both bank groups. I find no statistically significant change in this sensitivity for either group when the economy transitions into a low-interest-rate environment, and the estimated changes in sensitivity are not significantly different across groups. However, I find that banks with greater deposit market power increase the maturity of their non-loan assets more following the 2020 rate cut. These findings provide direct empirical support that banks increase the maturity of their assets—particularly bonds—to offset the increased interest rate sensitivity of deposit spread when the economy enters low-interest-rate environments.

Because invested capital generates cash flows with different interest rate risks compared to banks' deposit business, the leverage ratio can affect the interest rate risk of their overall earnings, and thus influence asset maturity choices. However, I find that banks' leverage ratios have remained stable over the past two decades, with minimal cross-sectional variation, making them unlikely to drive the variation in asset maturity. Furthermore, I demonstrate that neither non-interest income nor non-interest expenses—nor their combined cash flow—is exposed to

interest rate risk in either high- or low-interest-rate environments. Therefore, these sources of cash flows are also unlikely to explain the variation in asset maturity.

Related Literature. This paper contributes to several strands of the banking literature. First, a long line of research documents that banks' earnings are highly stable and exhibit little correlation with movements in interest rates (e.g., Flannery (1981); Flannery (1983); English (2002); Drechsler, Savov, and Schnabl (2021); Begenau and Stafford (2022)). Recently, Drechsler, Savov, and Schnabl (2021) and DeMarzo, Krishnamurthy, and Nagel (2024) provide evidence that banks choose asset maturities to align the interest rate sensitivity of asset-side cash flows with those on the liability side. DeMarzo, Krishnamurthy, and Nagel (2024) conjectures that this behavior aims to maintain stable earnings, although the focus of their research is not on why banks aim to maintain stable earnings. I contribute to this discussion by demonstrating that regulatory capital requirements can incentivize banks to stabilize their earnings, providing a microfoundation for this observed earnings stability.

This paper builds on the seminal work of Drechsler, Savov, and Schnabl (2017), which demonstrates that banks' deposit market power allows them to maintain deposit betas below one, and that deposit betas are negatively associated with banks' deposit market power. I extend this literature by incorporating insights from the macroeconomics literature (Ulate, 2021), which finds that ZLB makes deposit rate pass-through more sluggish, i.e. smaller deposit betas, in low-interest-rate environments. I show that this reduction in deposit betas, in turn, incentivizes banks to increase their assets maturities in low-interest-rate environments.

This paper also contributes to the literature on monetary policy and financial stability. Since Rajan (2005) proposed that low interest rates induce financial institutions to 'reach for yield' by investing in higher-yield and riskier assets, a large literature has confirmed this behavior across various financial institutions, including insurance companies (Becker and Ivashina, 2015), money market mutual funds (Di Maggio and Kacperczyk, 2017), corporate bond mutual funds (Choi and Kronlund, 2018), and commercial banks (Bubeck, Maddaloni, and Peydró,

2020). Prior studies have generally attributed this phenomenon to agency friction or accounting considerations. This paper proposes another potential mechanism: banks reach for yield in the low-interest-rate environment due to their risk management strategies, which dictate them to maintain stable earnings.

These results thus complement prior studies that show low interest rates increase banks' credit risk-taking (e.g., Maddaloni and Peydró (2011); Jiménez, Ongena, Peydró, and Saurina (2014); Martinez-Miera and Repullo (2017); Heider, Saidi, and Schepens (2019)) and decrease banks' loans growth (e.g. Ulate (2021); Wang, Whited, Wu, and Xiao (2022); Abadi, Brunnermeier, and Koby (2023); Eggertsson, Juelsrud, Summers, and Getz Wold (2024); Wang (2025)) by highlighting that it also leads to an increase in banks' interest rate risk. As a result, this paper provides insight into the significant losses experienced by banks following the rapid increase in the Fed funds rate in 2022 and contributes to the growing literature on the 2023 US banking turmoil (e.g., Haddad, Hartman-Glaser, and Muir (2023); Jiang, Matvos, Piskorski, and Seru (2024); Granja, Jiang, Matvos, Piskorski, and Seru (2024); Drechsler, Savov, Schnabl, and Wang (Forthcoming)).

The remainder of this paper is structured as follows. Section 2 presents the model. Section 3 describes the data and presents timeseries evidence. Section 4 discusses the diff-in-diff analysis. Section 5 provides further empirical analysis. Finally, Section 6 concludes the paper. All proofs and additional theoretical analyses are presented in Appendix A, and additional empirical analyses can be found in Appendix B.

2 The Model

In this section, I present a model that features common ingredients in the banking literature, including decreasing returns to scale in lending, capital requirements, deposit market power, and the ZLB on deposit rates. Using this model, I derive the following testable implications that will guide my empirical analysis in the subsequent sections. First, capital requirements and

decreasing returns to scale in lending give banks an incentive to maintain stable earnings, which they achieve by choosing asset maturities according to their deposit betas. Second, deposit betas are lower in low-interest-rate environments, which incentivize banks to extend asset maturities in these environments. Third, deposit betas tend to decline more for banks with greater deposit market power in low-interest-rate environments, which implies that these banks will increase their asset maturities more in these environments.

2.1 The Setting

The model has three dates $t = 0, 1, 2$. At $t = 0$, there is a safe short-term asset that pays an interest rate r_0 and a safe long-term asset whose return will be derived below. At $t = 1$, there is a safe short-term asset that pays an interest rate r_1.

Banks: There exists a continuum of banks that are managed by risk-neutral bankers. At $t = 0$, each bank allocates its initial capital E_0 in short term asset S_0 and long-term asset Q.[7] For simplicity, I assume that banks' other business operations in the initial period, including deposit-taking and lending, have occurred prior to the decision to invest in short-term and long-term assets, and that all profits and losses from these operations are included in the initial capital.[8]

At $t = 1$, each bank raises deposits D, grant loans L, and invest in the short-term asset S_1 in order to maximize its equity value E_2 at $t = 2$.[9] The bank faces a downward-sloping loan demand curve and upward-sloping deposit demand curves, as specified below.

At $t = 2$, all assets and liabilities are due, and all profits are realized. Banks will be liquidated, and the risk-neutral bankers will consume all net proceeds. At any dates, banks are allowed to borrow and lend at the risk-free rate r_t.

[7] Because banks are symmetric, I omit the superscript denoting bank j to simplify the notation, unless its inclusion is necessary.

[8] Incorporating these decisions into the model is straightforward. However, as will become clear, including them does not alter the core mechanism but would introduce unnecessary complexity.

[9] In the theoretical discussion, I do not explicitly differentiate between capital and equity, using the terms interchangeably.

Financial Frictions: Banks are subject to a forward-looking capital requirement, aligning with recent literature such as Brunnermeier and Koby (2019), Wang et al. (2022), and Eggertsson et al. (2024).

$$E_2 \geq \gamma L \tag{1}$$

where γ is the capital requirement for loans. This form of capital requirement captures two important features of the current regulatory framework. First, capital requirement is risk-weighted, with bonds have much lower risk weights compared to loans. Second, the capital is accumulated through banks' profits (i.e., earnings). In this research, I focus on the scenario where the capital constraint is binding. For simplicity, I assume the cost of raising outside capital is prohibitively high, so that the bank cannot raise outside capital. It can be shown that the results hold as long as raising capital is subject to financial frictions, such as adverse selection, which impose increasing costs on raising capital.

The Deposit Market: Banks offer differentiated deposit products in the deposit market. At $t = 1$, there will be a measure one of atomistic households with wealth $y = (1 + r_0)y_0$ that will enter the deposit market.[10] These households live until date $t = 2$ but have no income at $t = 2$, so they must save a portion of their wealth at $t = 1$ to finance future consumption. As banks offer differentiated deposit products, households make decisions in two stages: first, they choose a bank with which to deposit their funds; second, they determine the amount to consume and save.

After choosing bank j, a household's consumption and saving decisions are characterized by the following:

$$U(c_1, c_2) = \ln(c_1) + \beta \ln(c_2) \tag{2}$$

$$c_1 + d + m = y \tag{3}$$

[10] As will be shown, this assumption helps prevent the initial risk-free rate from mechanically influencing banks' asset maturity through its effect on the leverage ratio. This occurs because deposits are raised at time $t = 1$, while banks invest their initial capital at time $t = 0$. Consequently, a higher initial risk-free rate reduces banks' leverage ratio at $t = 1$. Although this unintended channel aligns with the direction of my main results regarding banks' optimal asset maturity, it is inconsistent with the stable leverage ratios observed in the data.

$$(1+r_D^j)d+m=c_2 \tag{4}$$

Here, c_1 and c_2 denote the household's consumption at dates $t=1$ and $t=2$, respectively. The household can choose to save at the selected bank, which offers a deposit rate r_D^j, or hold savings as cash m, which earns zero interest.

Importantly, the household's utility also depends on the bank it chooses. This first-stage decision is modeled using a stochastic utility framework:

$$v^j = v(r_D^j)+\mu\varepsilon^j \tag{5}$$

where $v(r_D^j)$ represents the indirect utility derived from the household's optimal consumption and saving decisions. The parameter μ is a positive constant, and ε^j is assumed to be independently and identically distributed according to a Type I extreme value distribution with zero mean and unit variance. We assume $\mu<\beta$, which implies that deposits across different banks are substitutes. The term $\mu\varepsilon^j$ captures horizontal differentiation among banks. As μ increases, consumer preferences become more heterogeneous, making households less sensitive to the deposit rate and more responsive to bank-specific characteristics such as service quality and geographic proximity. In other words, a higher μ implies greater deposit market power for banks.

Loans Demand: Each bank faces a downward-sloping loan demand curve, expressed as:

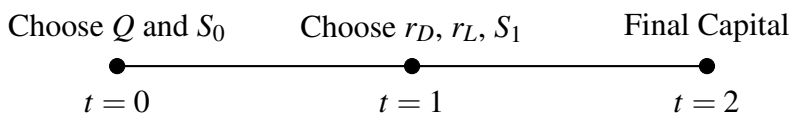
$$L(r_L)=a-b(r_L-r_1) \tag{6}$$

In Appendix A.1, I provide a microfoundation for this demand curve, where the bank's information advantage in monitoring local borrowers grants it market power in the loan market, leading to the downward-sloping loan demand curve. Since the focus of this paper is on the role of capital requirement and banks' deposit business, I will leave the detailed discussion for the loans demand in the appendix.

Interest Rate Shocks and the Long-term Rate: The only uncertainty in this economy is the interest rate shock at $t = 1$. It is assumed that the future risk-free rate r_1 follows a conditional probability distribution $f(r_1|r_0)$, with $\bar{r}_1 = E(r_1|r_0)$. This distribution is discrete, with equal probabilities assigned to three outcomes: $\bar{r}_1$, $\bar{r}_1 + \Delta$, and $\bar{r}_1 - \Delta$.[11] Additionally, it is assumed that higher current risk-free rate will imply higher expected future risk-free rate, i.e. if $r_0' > r_0$, then $E(r_1|r_0') > E(r_1|r_0)$.

In this model, it is assumed that the expectation hypothesis holds, i.e., the payoff at $t = 2$ from investing in the long-term asset is equivalent to investing and rolling over in short-term assets. This implies that the return from investing in the long-term asset at $t = 2$ is $1 + r_Q = (1 + r_0)(1 + \bar{r}_1)$.

Timeline and Bankers' Problem. To conclude, the timeline can be summarized as follows.



1. At $t = 0$, each bank chooses between short-term asset S_0 and long-term asset Q.

2. At $t = 1$, the interest rate shock r_1 realizes. Bankers set deposit rates to take new deposit D, and set loans rate to grant loans L, taking into account the capital constraint. If needed, they can borrow and lend at risk-free rate.

3. At $t = 2$, all assets and liabilities are due, and banks are liquidated.

Bankers' problem can be summarized as the following.

$$\max_{Q, r_L, r_D, S_1, S_0} \mathbf{E}_0(E_2|r_0) \tag{7}$$

$$E_2 = (1 + r_L)L + (1 + r_1)S_1 - (1 + r_D)D \tag{8}$$

$$L + S_1 = D + E_1 \tag{9}$$

[11] In Appendix A.3, I demonstrate that the results remain robust with asymmetric interest rate shocks.

$$E_1 = \frac{(1+r_0)(1+\bar{r}_1)}{1+r_1}Q + (1+r_0)S_0 \tag{10}$$

$$Q + S_0 = E_0 \tag{11}$$

$$E_2 \geq \gamma L \tag{12}$$

Equations 9 and 11 represents banks' balance sheet identity at $t = 1$ and $t = 0$. Equations 8 and 10 summarize banks' capital at $t = 2$ and $t = 1$.[12] Equation 12 is the capital constraint.

2.2 Capital Requirements, Earnings Stability, and Banks' Asset Maturity Choices

In this section, I solve banks' optimization problems using a general deposit demand function to illustrate how capital requirements incentivize banks to maintain stable earnings. Since banks' deposit profits, defined as the deposit spread multiplied by deposit amounts, fluctuate with the risk-free rate, these fluctuations incentivize banks to choose asset maturities that help stabilize earnings.

I solve the model using backward induction. At $t = 1$, the bank chooses deposit rate r_D and loans rate r_L (hence short-term securities S_1) to maximize its equity value at $t = 2$, given by Equations 8. Solving for S_1 from the balance sheet identity in Equation 9 and substituting it into Equation 8 yields:

$$E_2 = (r_L - r_1)L + (r_1 - r_D)D + (1+r_1)E_1 \tag{13}$$

The term $\pi_L = (r_L - r_1)L$ represents the profit from lending, which equals the spread between the loan rate r_L and the risk-free rate r_1, multiplied by the total amount of loans L. The term $\pi_D = (r_1 - r_D)D$ represents the profit from deposit-taking, calculated as the spread between the risk-free rate r_1 and the deposit rate r_D, multiplied by the volume of deposits D. Finally, the

[12] To derive equation 10, notice that investing Q in long-term asset will generate return $(1+r_0)(1+\tilde{r}_1)Q$ at $t = 2$. Therefore, the value of this long-term asset position at $t = 1$ is $\frac{(1+r_0)(1+\tilde{r}_1)}{1+r_1}Q$.

term $(1+r_1)E_1$ reflects the return on the bank's equity invested in short-term securities.

Notice that the bank's deposit rate-setting decision is independent of its investment decisions, which implies that the optimal deposit profit can be expressed as follows:[13]

$$\pi_D^*(r_1) = \max_{r_D}(r_1 - r_D)D \tag{14}$$

Given the deposit profit, the bank sets loans rate to maximize its final equity, i.e.,

$$E_2 = \max_{r_L}(r_L - r_1)L + (1+r_1)E_1 + \pi_D^*(r_1) \tag{15}$$

$$s.t. \quad E_2 \geq \gamma L$$

Which gives the optimal final equity as a function of the $t = 1$ risk-free rate and the $t = 1$ equity level E_1, i.e. $E_2(E_1, r_1)$. Because the choice of long-term asset could affect the level of $t = 1$ equity level E_1, the bank chooses the amount of long-term asset at $t = 0$ to maximize its expected final equity, i.e.,

$$\max_{Q} \quad \mathbf{E}_0[E_2(E_1, r_1)|r_0] \tag{16}$$

$$E_1 = \frac{(1+r_0)(1+\bar{r}_1)}{1+r_1}Q + (1+r_0)S_0 \tag{17}$$

$$Q + S_0 = E_0 \tag{18}$$

Solving this problem and normalizing the optimal amount of long-term assets by the initial capital yields the optimal asset maturity level q^* and the following results:

Proposition 1. *Banks' optimal asset maturity q^* is given by:*

$$q^* = \frac{Q^*}{E_0} = 1 + \frac{1}{\bar{E}_1}\left[\frac{\pi_D^*(\bar{r}_1 + \Delta) - \pi_D^*(\bar{r}_1 - \Delta)}{2\Delta}\right] \tag{19}$$

[13] It is straightforward to show that this independence holds if banks are net investors in short-term risk-free securities or if they can borrow at the risk-free rate.

It follows that:

1. *The optimal asset maturity minimizes fluctuations in banks' earnings and final capital E_2 across different states, thereby also minimizing fluctuations in lending.*

2. *The optimal asset maturity increases with the interest rate sensitivity of deposit profits, measured by $\frac{\pi_D^*(\bar{r}_1+\Delta)-\pi_D^*(\bar{r}_1-\Delta)}{2\Delta}$.*

3. *The optimal asset maturity decreases with the bank's capital $\bar{E}_1$.*[14]

To understand this result, recall that the expected final capital is given by:

$$\mathbf{E}[E_2(E_1,r_1)|r_0] = \mathbf{E}[\pi_D^* + \pi_L + (1+r_1)E_1|r_0] \tag{20}$$

This expression consists of three components: the expected profit from deposits, $\mathbf{E}_0(\pi_D^*)$; the expected profit from lending, $\mathbf{E}_0(\pi_L)$; and the expected return on capital investment, $\mathbf{E}_0[(1+r_1)E_1] = (1+r_0)(1+\bar{r}_1)E_0$. Because the expected return from capital investment and the expected deposit profit are determined by the level of risk-free rate, the only component banks can influence is the expected profit from lending.

When the capital constraint is binding, lending exhibits decreasing returns to scale, which implies that the profit from lending is an increasing and concave function of the lending volume.[15] Consequently, due to Jensen's inequality, fluctuations in lending are ex ante undesirable for banks. Given the capital constraint, the amount the bank can lend is determined by its capital, which accumulates through earnings (i.e., overall profit). Therefore, fluctuations in earnings are also undesirable, providing the bank with a strong incentive to maintain stable earnings.[16]

[14] Here, $\bar{E}_1 = (1+r_0)E_0$ represents the capital at $t=1$ when the interest rate shock is at its expected level, which is also the level of capital at $t=1$ if the bank does not invest in long-term assets. The normalization is based on the capital at $t=1$ because deposit-taking occurs at $t=1$.

[15] It can be shown that profit from lending is

$$\pi_L(L) = [r_L(L) - r_1]L = \frac{a-L}{b}L$$

which is increasing and concave in the lending volume L when $L < \frac{a}{2}$, i.e., when the capital constraint is binding.

[16] The intuition is similar to the insight from the literature of corporate risk management (e.g. Froot, Scharfstein,

In this model, there are two sources of earnings fluctuations. First, increase in risk-free rate increases the profit from deposits and the investment return of capital, where the first relationship follows from results related to banks' deposit market power that will be discussed in the next section. Second, the profit of investing in the long-term asset for amount Q, net of funding cost (i.e., $(1+r_0)(\bar{r}_1 - r_1)Q$), decreases when risk-free rate increases. As a result, banks will choose the amount of long-term asset that minimize the fluctuation in earnings and capital, thereby also minimizing the fluctuation in lending. Indeed, in Appendix A.1, I show that the amount of long-term assets that minimizes fluctuations in lending is also the amount of long-term assets that maximizes the expected final equity.

When the sensitivity of deposit profit increases, a given interest rate shock leads to larger fluctuations in deposit profit. As a result, the bank is incentivized to choose a higher asset maturity to stabilize its earnings. Meanwhile, holding the size of the bank's deposit business constant, higher capital implies that the proportion of earnings derived from deposits decreases. Consequently, the bank needs to hold fewer long-term assets to hedge against fluctuations in this component of earnings—that is, it will choose a lower asset maturity.

As a result of these asset-liability managements, banks' earning should have little interest rate risk exposure, as summarized in the following empirical prediction:

Empirical Prediction 1. *The correlation between banks' earnings and the risk-free rate is close to zero.*

Notice that this is a well-established empirical result in the literature, as discussed in the literature review. In Section 3.4, I verify this result and demonstrate that it holds both when risk-free rates are relatively high or in low-interest-rate environments.

and Stein (1993)), where firms with financial constraint have incentive to stabilize earnings if there are diminishing returns on investment.

2.3 Deposit Betas and Asset Maturity Choices Under Varying Interest Rates and Deposit Market Power

In this section, I show that the interest rate sensitivity of deposit profit is negatively related to the deposit beta—that is, the pass-through rate from the risk-free rate to the deposit rate. I then analyze both the time-series and cross-sectional variation in deposit beta, which in turn drives the time-series and cross-sectional variation in banks' optimal asset maturity.

I begin the discussion with the household's problem. In Appendix A.2, I solve the household's optimization problem and incorporate the distributional assumption for the stochastic utility component. This leads to a deposit demand function for bank j, which can be expressed as:

$$D^j = \begin{cases} (\frac{1+r_D^j}{1+r_D})^{\frac{\beta}{\mu}} \frac{\beta}{1+\beta} y, & \text{if } r_D^j \geq 0 \\ 0, & \text{if } r_D^j < 0 \end{cases} \tag{21}$$

Where $1 + r_D = (\int_0^1 (1 + r_D^r)^{\frac{\beta}{\mu}} dr)^{\frac{\mu}{\beta}}$ represents the aggregate deposit rate. Plugging this deposit demand function into the bank's deposit profit maximization problem (Equation 14), we derive a simple piece-wise optimal deposit rate setting rule that each bank follows in equilibrium:

$$1 + r_D^* = 1 + r_D^j = \begin{cases} 1 & \text{if } r_1 < \frac{\mu}{\beta} \\ \frac{\beta}{\beta+\mu}(1 + r_1) & \text{if } r_1 \geq \frac{\mu}{\beta} \end{cases} \tag{22}$$

In Figure 2, the optimal deposit rate function is shown as the solid blue line. When the risk-free rate is sufficiently high ($r_1 > \frac{\mu}{\beta}$), the deposit rate is a markdown of the risk-free rate, where a one-basis-point increase in the risk-free rate raises the deposit rate by $\frac{\beta}{\beta+\mu}$. When the risk-free rate is below a threshold ($r_1 \leq \frac{\mu}{\beta}$), the deposit rate is zero because the ZLB constrains banks' ability to reduce it further.

The corresponding equilibrium deposit quantity is:

$$D^* = D^j = \frac{\beta}{1+\beta} y \tag{23}$$

Note that this model implies a deposit quantity that remains constant regardless of the level of the risk-free rate at date $t = 1$, which aligns with empirical observations that deposit quantities tend to remain relatively stable in consecutive periods.

Incorporating the optimal deposit rate setting rule into the deposit profit function, we have

$$\frac{\pi_D^*(\bar{r}_1 + \Delta) - \pi_D^*(\bar{r}_1 - \Delta)}{2\Delta} = D(1 - \beta^D) \tag{24}$$

where the deposit beta $\beta^D = \frac{r_D^*(\bar{r}_1+\Delta) - r_D^*(\bar{r}_1-\Delta)}{2\Delta}$ denotes the average pass-through rate from the risk-free rate to the deposit rate between the levels $\bar{r}_1 + \Delta$ and $\bar{r}_1 - \Delta$. This result is intuitive: since deposit profit equals the deposit spread multiplied by the deposit quantity, the interest rate sensitivity of deposit profit is inversely related to the interest rate sensitivity of the deposit rate, i.e., the deposit beta.

From the optimal deposit rate-setting rule, it is straightforward to derive the cross-sectional and time-series variation in the deposit beta, as summarized in the following proposition:

Proposition 2. *The relationship between banks' deposit betas, risk-free rate levels, and deposit market power can be summarized as follows:*

1. *When $E(r_1|r_0) - \Delta > \frac{\mu}{\beta}$, banks' deposit beta β^D remains constant with respect to the risk-free rate r_0.*

2. *When $E(r_1|r_0) \in \left[\frac{\mu}{\beta} - \Delta, \frac{\mu}{\beta} + \Delta\right]$, banks' deposit beta β^D increases with the level of the risk-free rate r_0.*

3. *When $E(r_1|r_0) + \Delta \leq \frac{\mu}{\beta}$, the deposit beta is zero for all banks.*

4. *When $E(r_1|r_0)+\Delta > \frac{\mu}{\beta}$, banks' deposit beta β^D is negatively related to their deposit market power μ.*

There are three regimes for deposit betas. In the normal regime, where the risk-free rate is sufficiently high, banks do not anticipate scenarios in which they would need to set the deposit rate to zero at $t = 1$. Consequently, the deposit beta is constant and is determined primarily by a bank's deposit market power, which is consistent with the findings in Drechsler, Savov, and Schnabl (2017). In the "ultra-low" interest rate regime, characterized by a low initial risk-free rate r_0 and low volatility of the risk-free rate Δ, banks expect to always set the deposit rate to zero at $t = 1$, resulting in a deposit beta of zero for all banks.

In the low interest rate regime, where the initial risk-free rate r_0 is lower than in the normal regime but higher than in the ultra-low regime, a bank sets the deposit rate to zero if the realized risk-free rate is sufficiently low ($r_1 = E(r_1|r_0) - \Delta$) and a positive deposit rate when the realized risk-free rate is sufficiently high ($r_1 = E(r_1|r_0) + \Delta$). As a result, its deposit beta falls below the "normal" level but remains above zero (see the dashed line in Figure 2 for an illustration, its slope represents the deposit beta when the expected future rate is $\bar{r}_1 = E(r_1|r_0)$).

Overall, Proposition 2 indicates that (1) banks' deposit betas increase with the risk-free rate when the risk-free rate is low and remains constant when the risk-free rate is sufficiently high; and (2) banks' deposit betas decrease with their deposit market power. Combining Equation 19 and Expression 24, we derive the relationship between banks' optimal asset maturity, their deposit betas, and their deposit market power:

Proposition 3. *A bank's optimal asset maturity can also be expressed as:*

$$q^* = 1 + \lambda(1 - \beta^D) \tag{25}$$

Where $\lambda = \frac{\beta y_0}{(1+\beta)E_0}$ denotes the bank's leverage ratio. It follows that:

1. *A bank's optimal asset maturity q^* is negatively related to its deposit beta β^D.*

2. *When* $E(r_1|r_0) \in \left[\frac{\mu}{\beta} - \Delta, \frac{\mu}{\beta} + \Delta\right]$, *a bank's optimal asset maturity* q^* *decreases with the level of the risk-free rate* r_0. *Otherwise,* q^* *remains constant with respect to the risk-free rate* r_0.

3. *When* $E(r_1|r_0) \in \left[\frac{\mu}{\beta} - \Delta, \infty\right)$, *a bank's optimal asset maturity* q^* *is positively related to its deposit market power* μ.

These results are intuitive. When the deposit beta is smaller, a given interest rate shock causes smaller fluctuations in the deposit rate and larger fluctuations in the deposit spread (the same to the deposit profit), which incentivizes the bank to choose longer asset maturities to stabilize earnings. Because deposit betas decrease when the risk-free rate is low, banks' optimal asset maturity will be higher. Similarly, since banks with greater deposit market power tend to have lower deposit betas, their optimal asset maturity will also be higher.

Combining Proposition 2 and Proposition 3, we derive the following empirical predictions regarding the relationship between deposit betas and asset maturities across different interest rate environments and for banks with different level of deposit market power:

Empirical Prediction 2.

1. *In low-interest-rate environments, banks exhibit lower deposit betas and longer assets maturities, relative to periods when risk-free rates are high.*

2. *Banks with greater deposit market power exhibit lower deposit betas and longer asset maturities.*

In the subsequent sections, I provide empirical evidence supporting both predictions. Notably, the second prediction aligns with prior findings in the literature (Drechsler, Savov, and Schnabl, 2021).

Finally, by comparing the deposit beta and optimal asset maturity of a bank with greater deposit market power to those of a bank with lower deposit market power, we obtain the following results:

Proposition 4. *There exist two risk-free rate levels, r_0^H and r_0^L, with $r_0^H > r_0^L$, such that when the risk-free rate declines from r_0^H to r_0^L:*

1. *Deposit betas β^D decrease more for banks with greater deposit market power.*

2. *Optimal asset maturities q^* increase more for banks with greater deposit market power.*

This relationship can be intuitively illustrated with the following example: consider a bank with no deposit market power and another bank with some degree of deposit market power. For the bank with no market power, both the deposit beta and asset maturity remain fixed, regardless of changes in the risk-free rate. In contrast, as established in Proposition 3, the deposit beta decreases and the optimal asset maturity increases for the bank with deposit market power when the risk-free rate falls. Appendix A.2 provides a simple proof of this proposition.

Importantly, Proposition 4 establishes two difference-in-differences empirical predictions, summarized as follows:

Empirical Prediction 3. *Following an interest rate shock that shifts the economy from a relatively high risk-free rate environment to a low-interest-rate environment:*

1. *Deposit betas decline more for banks with greater deposit market power.*

2. *Asset maturities increase more for banks with greater deposit market power.*

In Section 4, I empirically confirm these predictions by showing that deposit betas decline more for banks with greater deposit market power in low-interest-rate environments. Moreover, I exploit the Fed funds rate cuts induced by the COVID-19 pandemic to demonstrate that banks' asset maturities increase more for banks with greater deposit market power in low-interest-rate environments.

2.4 Discussions

The Interest Rate Risk of Banks' Value. I have shown that capital requirements can incentivize banks to maintain stable earnings, which in turn prompts them to increase asset maturity

in low-interest-rate environments. Although this strategy can insulate banks from the interest rate risk associated with earnings, their overall valuation remains sensitive to interest rate fluctuations due to the discount rate effect. Consequently, an increase in interest rates reduces the present value of banks. This result aligns with empirical evidence. For example, English, Van den Heuvel, and Zakrajšek (2018) exploit exogenous fluctuations in interest rates induced by FOMC announcements and show that bank stock prices tend to decline when interest rates rise.

Run Risk. An important assumption of the model is that depositors remain with banks even when banks incur losses in their securities portfolios. If this assumption is relaxed—i.e., depositors may withdraw their funds if they believe banks are insolvent—then bank runs could happen, as discussed in Jiang et al. (2024) and Drechsler et al. (Forthcoming). In this paper, I study banks' interest rate risk management strategies and their implications for banks' asset maturity decisions across different interest rate environments, under the assumption that banks neglect run risk. This assumption is supported by the fact that bank runs were rare events and did not occur during the period between the 2007–2008 Global Financial Crisis and the 2023 banking turmoil.

Discussions to the Model Setup. This paper provides simple microfoundations for both the lending and deposit markets. However, the framework is flexible and can accommodate alternative microfoundations for these markets. The main results hold as long as the following three conditions are satisfied: (1) the bank's lending business exhibits decreasing returns to scale, (2) lending volume is constrained by capital requirements, and (3) deposit profit is an increasing and strictly concave function of the risk-free rate. The first two conditions are standard in the banking literature, while the third is an empirical regularity that I demonstrate with data.

Additionally, the framework implicitly assumes that profits from lending do not contribute interest rate risk to the bank's earnings. In practice, this assumption may not hold due to factors such as fluctuations in credit demand and changes in credit risk. However, as long as this interest

rate risk remains stable across high and low interest rate environments, the main result—that banks increase their asset maturity in response to heightened interest rate sensitivity of deposit profits when risk-free rates decline—continues to hold. In Section 4.4, I provide empirical evidence supporting this assumption.

Finally, one potential concern is that banks may use interest rate derivatives to stabilize their earnings, which is not incorporated into the model. However, this concern may be overstated in the context of the U.S. banking industry. Empirical studies (e.g., McPhail, Schnabl, and Tuckman (2023) and Granja et al. (2024)) show that banks' net positions in interest rate derivatives are minimal relative to their overall exposure to interest rate risk. One reason for this limited use of derivatives is the significantly higher risk weights assigned to them. In contrast, among the primary bonds held by banks, only agency MBS from Freddie Mac and Fannie Mae carry a risk weight of 20%, while other commonly held bonds, such as U.S. government securities, have a risk weight of 0%. As a result, bonds become natural tools for banks to maintain stable earnings, as using derivative would reduce their lending capacity.

3 Data and Timeseries Evidence

In the theoretical analysis, I demonstrate that in low-interest-rate environments, deposit betas will be smaller, which implies that the interest rate sensitivity of banks' deposit profits will be higher. To hedge this heightened risk and maintain stable earnings, banks optimally choose longer asset maturities. In this section, I empirically measure banks' asset maturity and deposit betas in both high- and low-interest-rate environments to provide an initial test of the theory.

First, I measure banks' asset maturity using their reported repricing maturity and demonstrate that banks significantly increased their asset maturity as the economy entered low-interest-rate environments and decreased their asset maturity when interest rates rose. Moreover, this pattern occurs for both large and small banks, and across all of banks' different asset classes. These results imply that banks increase the interest rate sensitivity of their asset-side cash flows

by holding assets with longer maturities in low-interest-rate environments.

Next, I measure both the bank-level and industry average deposit betas in high and low interest rate environments and demonstrate that the deposit beta decreases in low-interest-rate environments, indicating an increase in the interest rate sensitivity of banks' deposit profits in low-interest-rate environments.

Finally, I demonstrate that banks' earnings, as measured by return on assets, are not correlated with the risk-free rate, both in the high and low interest rate environments. Overall, these results provide empirical support that the decrease in deposit betas prompt banks to increase asset maturity in order to maintain stable earnings.

3.1 Data

To construct the dataset, I used multiple data sources.

Call Report. The primary dataset I am using is the Call Report, which is from the Federal Financial Institutions Examination Council (FFIEC). The data provides quarterly bank-level information on balance sheets and income statements for all U.S. commercial banks. In the next section, I will carefully discuss how I measure banks' asset maturity. For all analysis, I aggregated the relevant variables to the bank holding company level. Besides, I also follow the convention in the literature by limiting bank holding companies to those with assets exceeding $100 million at the end of 2018.

FDIC Bank Data. I enhanced the call report data by incorporating FDIC bank data to include additional bank characteristics. Beyond the details in the Call report, the FDIC also gathers critical bank attributes, such as insured deposit ratios and bank holding company IDs. These data can be merged with Call report data using the unique bank identifiers.

FDIC Summary of Deposits. I got the branch-level deposit data and annual branch location data from FDIC Summary of Deposits. The FDIC collects these information from all FDIC-insured institutions as of June 30 each year. This data can be merged to Call Report data using FDIC unique identifier. Importantly, the data set has information on branch characteristics such

as the parent bank, address, and geographic coordinates.

3.2 Measuring Banks' Asset Maturity

In this section, I empirically demonstrate that banks increase their asset maturity in low-interest-rate environments, while they gradually reduce their asset maturity as interest rates rise. Furthermore, I show that this pattern occurs for both large and small banks and across different asset classes. These results imply that the interest rate risk of banks' asset side cash flows increases in low-interest-rate environments.

Specifically, I measure banks' asset maturity by averaging their repricing maturity across asset classes, with the repricing maturity information sourced from banks' call reports. For fixed-rate contracts, banks report the remaining maturity. In the case of floating-rate contracts, they report the period between the filing date and either the next repricing date or the contractual maturity date, whichever occurs first. Thus, the repricing maturity reflects the period during which interest rates are contractually fixed, as opposed to the notional maturity of the asset.

However, banks report only the range of months (or years) remaining until the repricing maturity, without specifying the exact timing. To estimate the repricing maturity of each asset class, I follow the methods of English, Van den Heuvel, and Zakrajšek (2018) and Drechsler, Savov, and Schnabl (2021) to assign the midpoint of the respective repricing maturity range reported in the Call Report as the maturity for each asset category.[17]

I then calculate banks' asset maturity as the weighted-average maturity across their asset portfolio, i.e.

$$q_{i,t} = \sum_{j=1}^{N_1} \omega_{i,t}^{j} rm_{i,t}^{j} \tag{26}$$

Here, $q_{i,t}$ denotes the asset maturity for bank i at date t, $\omega_{i,t}^{j}$ is the portfolio weight for asset class j and $rm_{i,t}^{j}$ is the corresponding average repricing maturity for asset class j.

[17] For example, U.S. government securities reported as having a remaining maturity or next repricing date of over 3 years through 5 years were assumed to have a repricing maturity of 4 years, the midpoint of the (3,5] interval. In Appendix B.1, I provide a detailed discussion of each asset category and show that the sample covers more than 98% of liabilities and 95% of assets during most of the sample period, making it highly comprehensive.

Banks' asset maturity increases when the economy enters low-interest-rate environments and gradually declines as interest rates rise. As illustrated in Figure 1, the median asset maturity in the banking sector is negatively correlated with the Fed funds rates. It increased significantly when the Fed funds rates were near zero, both following the Global Financial Crisis and the Covid-19 pandemic. Conversely, it gradually decreased during periods of rising Fed funds rates, notably between 2004-2006, 2015-2019, and after 2022. As a result of the ultra-low Fed funds rate following the Global Financial Crisis and the Covid-19 pandemic, the median asset maturity in the banking sector rose from 3 years in 2007 to nearly 6 years in 2021, indicating a substantial increase in the interest rate risk of banks' asset.

To ensure these trends are not solely driven by small community banks, Figure 3 displays the median asset maturity separately for the 200 largest banks by 2018Q4 asset value, representing regional and national banks, and for all other banks.[18] As the figure illustrates, both groups follow very similar trends, indicating that the observed patterns in asset maturity are consistent across bank sizes.

The increase in asset maturity also happens across banks' asset classes. In Figure 4, I plot the median maturity for securities, single-family residential mortgage, and (other) loans.[19] As the figure illustrates, the asset maturity of each class increased when the Fed funds rate was near zero, with the rise being more significant for securities and single-family residential mortgages. These findings mitigate concerns that the increase in asset maturity is driven by borrowers seeking to lock in low borrowing rates. If this were the case, banks could simply securitize more mortgages and hold shorter-term securities.

[18] Regulators use a loose cutoff of $10 billion to distinguish regional and national banks. As of 2023Q4, 160 banks exceed this asset threshold, with an additional 16 banks near it. Since bank asset values can fluctuate around this level, I define the top 200 banks by asset size to include all banks that might be categorized as regional or national banks by regulators.

[19] Single-family residential mortgages are separated from other loans due to their liquid secondary market, which allows banks to sell them easily. Additionally, the capital requirement for these mortgages is the same as that for mortgage-backed securities (MBS), rather than other types of loans. At last, prior literature (e.g. Hurst, Keys, Seru, and Vavra (2016)) shows that banks' local market power doesn't affect the rates for single-family residential mortgage. As a result, single-family residential mortgage are more akin to securities. For this reason, I exclude single-family residential mortgages from the loan calculations in this paper.

Overall, these empirical findings confirm the theoretical prediction that banks increase their asset maturity in low-interest-rate environments. This is consistent with the broader literature, which shows that low interest rate environments prompt financial institutions to reach for yield.[20] However, in the context of this study, banks increase their asset maturity to increase the interest rate sensitivity of their asset-side cash flows, which is used to offset the increased interest rate sensitivity of their deposit profits. In other words, banks reach for yield due to their risk management strategy.

3.3 Banks' Deposit Betas in High- and Low- Interest Rate Environments

In this section, I show that banks' deposit betas are lower in low-interest-rate environments, which implies that banks' deposit profits become more sensitive to interest rate changes in these environments. Combined with the finding that banks increase their asset maturities in low-interest-rate environments, this evidence confirms the model's empirical prediction and provides initial evidence that banks increase their asset maturities to maintain stable earnings.

Following the method proposed by Drechsler, Savov, and Schnabl (2021), I estimate the deposit beta for each bank i in both (relatively) high and low interest rate environment using the following regression:

$$\Delta r_{it}^{COF} = \alpha_i + \eta_t + \sum_{\tau=0}^{3} \beta_{i,\tau}^{D} \Delta r_{t-\tau} + \varepsilon_{it} \tag{27}$$

where $\Delta r_{i,t}^{COF}$ is the change in bank i's cost of funds at date t, which is defined as bank i's total interest expense divided by its total liabilities at date t, Δr_t is the change in the Fed funds rate at date t, α_i and η_t denote the bank and time fixed effects, respectively. The deposit beta of bank i is the sum of the coefficients $\beta_i^D = \sum_{\tau=0}^{3} \beta_{i,\tau}^{D}$. Given that low-interest-rate environments lasted for 13 years, from 2009 to 2021, I used data from this period to estimate the deposit betas for low-interest-rate environments. To ensure a meaningful comparison, I estimated the

[20] These financial institutions include insurance companies (Becker and Ivashina, 2015), money market mutual funds (Di Maggio and Kacperczyk, 2017), corporate bond mutual funds (Choi and Kronlund, 2018), and the European banking industry (Bubeck, Maddaloni, and Peydró, 2020).

deposit betas for (relatively) high interest rate environment using data from the preceding 13 years, specifically from 1996 to 2008.

In Figure 5, I plot the estimated deposit betas for both (relatively) high- and low-interest-rate environments. As shown in the figure, the entire distribution of deposit betas shifts to the left as the economy transitions to low-interest-rate environments. The average estimated deposit beta during high-interest-rate environments is 0.365, closely aligning with the estimation results in Drechsler, Savov, and Schnabl (2021). However, in low-interest-rate environments, the average estimated deposit beta declines significantly to 0.248.

To formally test the hypothesis that banks' deposit betas decreased in low-interest-rate environments, I estimate the following regression using US banking data from 2002 to 2024:

$$\Delta r_{i,t}^{COF} = \alpha_i + \sum_{\tau=0}^{3} \beta_\tau^D \Delta r_{t-\tau} + \sum_{\tau=0}^{3} \gamma_\tau^D \Delta r_{t-\tau} \mathbb{1}(r_{t-\tau} \leq \tilde{r}) + \beta \text{Control}_{i,t} + \varepsilon_{i,t} \tag{28}$$

Here, $\mathbb{1}(r_{t-\tau} \leq \tilde{r})$ represents an indicator function that identifies whether the shock occurred in a low-interest-rate environment, where a low-interest-rate environment is defined as the Fed funds rate being below a specified threshold $\tilde{r}$. Controls include the bank's log of asset value and Tier-1 capital ratio. In the benchmark case, I use a threshold of 1.5%, and I show that the results are robust to other thresholds, such as 1% or 2%. In this formulation, β^D represents the average deposit beta in the banking industry when the risk-free rates are relatively high, and $\gamma^D = \sum_{\tau=0}^{3} \gamma_\tau^D$ captures the change in the average deposit beta in low-interest-rate environments. To account for the fact that the residuals may be correlated across time for a given bank and across banks for a given quarter, I double-cluster the standard errors at the bank and time levels.

As shown in column 2 of Table 1, the deposit beta significantly decreases when the economy transitions into low-interest-rate environments. Specifically, when the risk-free rate is relatively high, the estimated deposit beta is 0.370. However, in low-interest-rate environments, the deposit beta declines by 0.141, resulting in an deposit beta of 0.229 under these conditions.

A potential concern regarding this test is that the sample period spans over more than 20

years, during which regulations and market conditions may have changed, particularly following the financial crisis. To address this concern, I present results from a more recent period, 2014Q2 to 2024Q2, in column 3 of Table 1. As it shows, the estimation results remain consistent with those obtained from the full sample period. Besides, I repeat the analysis for the largest 200 banks in column 4 of Table 1 and also find consistent results. The deposit beta of these large banks are generally higher, which also aligns with previous literature.

Additionally, I re-estimate the regression with threshold values set at 1% and 2% in columns 5 and 6, respectively. In both cases, the deposit beta significantly decreases in the low-interest-rate environments. The theoretical model suggests that the estimated deposit beta with a 1% threshold should be smaller than with a 1.5% threshold, and the estimated deposit beta with a 2% threshold should be larger than with a 1.5% threshold.[21] This hypothesis is supported by the data: when defining a low-interest-rate environment as periods with Fed funds rates below 1%, the estimated deposit beta is 0.157 (= 0.381 - 0.224), which is less than the estimated deposit beta with a 1.5% threshold, i.e., 0.229. Conversely, when the threshold is set at 2%, the estimated deposit beta is 0.252 (= 0.372 - 0.120), which is greater than 0.229.

In conclusion, the empirical evidence suggests that banks' deposit betas are smaller in low-interest-rate environments, which confirms my model's empirical prediction. Notably, this result also aligns with banks' own assessments. For example, Citi Bank's 2020 10-K states, "At progressively higher interest rate levels, ... as Citi assumes it will pass on a larger share of rate changes to depositors (i.e., higher betas)." Similarly, in the November 2022 Senior Financial Officer Survey, bank managers generally indicated that the expected deposit beta for the next six months was higher than the realized deposit beta for the previous six months.

[21] Although I did not explicitly present this empirical prediction to conserve space, it follows that deposit betas increase with the risk-free rate in low-interest-rate regimes.

3.4 The Interest Rate Risk of Banks' Earnings

In this section, I demonstrate that banks' earnings is not correlated with interest rates, regardless in high or low interest rates environments. Specifically, I estimated banks' ROA beta, which captures the pass-through of the Fed funds rate on banks' return on assets (ROA, defined as net income divided by total assets), using the following regression model:

$$\Delta \text{ROA}_{i,t} = \alpha_i + \sum_{\tau=0}^{3} \beta_\tau^{ROA} \Delta r_{t-\tau} + \sum_{\tau=0}^{3} \gamma_\tau^{ROA} \Delta r_{t-\tau} \mathbb{1}(r_{t-\tau} \leq \tilde{r}) + \beta \text{Control}_{i,t} + \varepsilon_{i,t} \tag{29}$$

Here, the ROA beta in a (relatively) high interest rate environment is represented by $\beta^{ROA} = \sum_{\tau=0}^{3} \beta_\tau^{ROA}$, and it changes by $\gamma^{ROA} = \sum_{\tau=0}^{3} \gamma_\tau^{ROA}$ in a low interest rate environment.

The empirical findings extend previous literature by confirming that banks' earnings remain uncorrelated with interest rates in both high and low-interest-rate environments. As presented in the Table 2, the estimated ROA beta, without distinguishing between high and low interest rate environments, is 0.029 and statistically insignificant. When differentiating between these environments, the estimated ROA betas are 0.021 for high-interest-rate environments and 0.027 for low-interest-rate environments, with neither estimate being statistically significant. In Section B.2, I provide estimates of ROA betas at the bank level, which show that the average estimated ROA beta is also close to zero in both high- and low-interest-rate environments.[22]

Overall, the results in this section provide preliminary empirical support for the theoretical analysis, suggesting that in low-interest-rate environments, banks' deposit betas are lower—that is, the interest rate sensitivity of banks' deposit profits is higher. In response, banks increase the interest rate sensitivity of their asset-side cash flows by increasing asset maturities, thereby maintaining stable earnings.

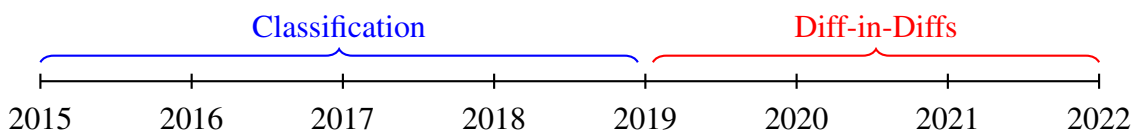
[22]The ROAs of some banks may be positively or negatively correlated with the Fed funds rate, as interest rate management at the bank level may not be perfect.

4 Difference-in-Differences Analysis

In this section, I provide additional evidence of the theory by utilizing the ultra-low Fed funds rate induced by the Covid-19 pandemic and the cross-sectional predictions of the theoretical model. Specifically, the model predicts that when the economy enters low-interest-rate environments, the deposit beta will be smaller, with a larger drop for banks with greater deposit market power. This implies that in such scenario, the interest rate sensitivity of banks' deposit profits will rise, with a larger rise for banks with higher deposit market power. Consequently, all banks tend to increase their asset maturity in low-interest-rate environments, but those with higher deposit market power will increase their asset maturity more.

To test the implications, I implement a mixture model to classify banks into high and low deposit market power groups based on their average deposit pass-through during the 2015–2018 monetary policy tightening period. The classification method is based on the theoretical insight that banks with greater deposit market power will increase deposit rates more slowly in response to rising Fed funds rates, when banks' deposit rate-setting decisions are independent of their investment decisions.

Using these classifications, I demonstrate that the deposit beta decreases for both groups of banks in low-interest-rate environments, but it declines more for banks with greater deposit market power. This suggests that the interest rate sensitivity of deposit profits increased for both groups, with a greater increase for banks with greater deposit market power. Supporting the theory, I find that both groups of banks increased their asset maturity following the ultra-low Fed funds rate of 2020Q2, but banks with higher deposit market power increased their asset maturity significantly more. To clarify how the samples are used, I summarize the sample periods for the classification and Diff-in-Diffs analyses in the following timeline:



In this analysis, one assumption is that the interest rate sensitivity of banks' lending profits does not change when the economy enters low-interest-rate environments. To empirically validate this assumption, I estimate this interest rate sensitivity for two groups of banks and find no significant change in this sensitivity for either group when the economy enters low-interest-rate environments.

Furthermore, the estimated changes in this sensitivity for the two groups are not significantly different, indicating that, even if the insignificant results at the group level are due to imprecise estimation, the change in this sensitivity is unlikely to differ between groups. However, I find that banks with greater deposit market power increase the maturity of their non-loan assets more, providing direct evidence that banks extend their asset maturity to stabilize earnings.

4.1 Classifying Banks into Deposit Market Power Groups

When banks' optimal deposit rates depend only on the risk-free rate and their deposit market power, banks with greater deposit market power will increase their deposit rates more slowly in response to rising Fed funds rates. In other words, for a given monetary tightening cycle, banks with greater deposit market power will have a smaller estimated β_i in the following bank level regression:

$$r^d_{i,t} = \xi_i + \beta_i r_t + \varepsilon_{i,t} \tag{30}$$

Here, $r^d_{i,t}$ is the deposit rate for bank i at date t and ξ_i is the bank-i specific intercept. Based on this observation, one potential method is to estimate Equation 30 using data from 2015-2018 monetary policy tightening period and categorize banks into those with estimated β_i values greater than the median and those with estimated β_i values smaller than the median. However, this classification procedure can introduce bias into the later diff-in-diff analysis. First, the limited observations available for each bank inevitably result in imprecise estimations of β_i. Second, and more importantly, classifying banks based on the median estimated β_i overlooks

the data's underlying cluster structure.[23]

To address these issues, I apply a mixture model approach, which is a well-established method in statistics and machine learning literature for clustering, to classify banks into different deposit market power groups. In Appendix B.4, I repeat the main analysis in this section using a sample where classification is based on bank-level estimation and find results consistent with those presented here. However, the magnitude of the estimated differences between groups is smaller than the results presented here, aligning with the idea that this classification procedure is less precise.

Bank Classification: Integrating the Economics Model with the Mixture Model Approach. Specifically, I assume the existence of K different deposit market power groups, each characterized by a unique β_k, where $k \in \{1, 2, \ldots, K\}$. Each bank i belongs to one of these groups, indicated by a latent variable z_i. This latent variable z_i is K-dimensional, with each dimension z_{ik} being a binary $0-1$ variable such that $\sum_{k=1}^{K} z_{ik} = 1$. Specifically, if $z_{ik} = 1$, bank i's deposit rates are assumed to follow the fixed-effect regression:

$$r_{i,t}^{d} = \xi_i + \beta_k r_t + \varepsilon_{i,t}$$

Within this framework, the log likelihood function given the data will be,

$$\ln p(\{\boldsymbol{r_i^d}\}_{i=1}^{N} | \theta) = \sum_{i=1}^{N} \ln \sum_{k=1}^{K} [\pi_k \mathcal{N}(\boldsymbol{r_i^d} | \beta_k \boldsymbol{r} + \xi_i \boldsymbol{e}, \Sigma_k)]$$

where $\boldsymbol{r}_i^d$, $\boldsymbol{e}$, and $\boldsymbol{r}$ are $T \times 1$ vectors, $\boldsymbol{e} = [1, 1, ..., 1]'$, $\boldsymbol{r}_i^d = [r_{i,1}^d, r_{i,2}^d, ..., r_{i,T}^d]'$, $\boldsymbol{r} = [r_1, r_2, ..., r_T]'$, and $p(z_{ik} = 1) = \pi_k$. θ denotes the set of all adaptive parameters, namely π_k, β_k, Σ_k, and ξ_i.

In order to maximize this likelihood function, I follow the statistics and machine learning

[23] In other words, there is no reason to assume an equal distribution of high and low deposit market power banks. This is problematic because the classification in this section seeks to uncover a stable characteristic that persists over relatively long periods. To verify this stability, I apply the same classification method to the post-2022 monetary policy tightening cycle and compare the results. Imprecise classification can significantly bias the findings in this subsequent analysis.

literature to apply the Expectation-Maximization (EM) algorithm (Dempster, Laird, and Rubin (1977), McLachlan and Krishnan (2007)). To avoid further complication, I leave the technical details in the appendix B.3. After estimating the model, I assign bank i to deposit market group based on its posterior probability distribution[24]:

$$\gamma_{ik} = E[z_{ik}] = p(k|\{\boldsymbol{r}_i^d\}_{i=1}^N, \theta) = \frac{\pi_k \mathcal{N}(\boldsymbol{r}_i^d|\beta_k \boldsymbol{r} + \xi_i \boldsymbol{e}, \Sigma_k)}{\sum_j \pi_j \mathcal{N}(\boldsymbol{r}_j^d|\beta_j \boldsymbol{r} + \xi_i \boldsymbol{e}, \Sigma_j)} \tag{31}$$

K=2 Case. In order to classify banks into high and low deposit market group, I estimate the mixture model with $K = 2$ and assign each bank to a group based on its posterior probability distribution. The estimated values of β_k are 0.070 and 0.283, respectively. I label banks with low estimated β_k as the high deposit market power banks and banks with high estimated β_k as the low deposit market power banks.

This method effectively separates banks into two groups. As Figure 6 shows, the posterior probabilities for a bank to be high deposit market power bank are concentrated to the region near 0 or 1, implying that banks behave very differently when setting deposit rates. I then label banks with this posterior probability greater than 0.8 as high deposit market power group and banks with this posterior probability smaller than 0.2 as the low deposit market power group. As a comparison, all banks with bank-level estimated β_i below median are classified as high deposit market power bank by the mixture model. In addition, it captured another 494 high deposit market power banks that have bank-level estimated β_i above the median.

To illustrate the classification results intuitively, I plotted the deposit rates against Fed funds rates for two groups of banks during the 2015-2018 monetary tightening period in figure 7. As shown in the figure, high deposit market power banks started to increase their deposit rates at a higher rate and increased deposit rate slower after they started to increase their deposit rates, which is consistent with the theoretical predictions.

Importantly, this method captures a relatively stable characteristic of the banks. To illustrate

[24]This follows by applying Bayes' theorem.

this, I first plotted the deposit rates against the Fed Funds rate for two groups of banks during the 2022Q1–2024Q2 tightening period. As shown in Figure 8, high deposit market power banks began increasing their deposit rates later and did so at a slower pace during this period, which indicates that the classification predicts the future behavior of banks well. Additionally, I applied the same algorithm to classify banks using the 2022Q1–2024Q2 sample period and compared the classification results with those from the 2015–2018 period. Despite the smaller sample size for the 2022Q1–2024Q2 period, 89% of the banks classified as having high deposit market power were also classified as having high deposit market power using the 2015–2018 period. These results suggest that deposit market power is a stable characteristic of banks that does not change significantly over short periods.

Group Characteristics. To understand other differences among banks across groups, I summarized the relevant bank characteristics as of 2018Q4 in Table 3. Several features arise from the table.

First, the characteristics of banks' liabilities align well with the theory presented in this paper and the existing literature (e.g., Drechsler, Savov, and Schnabl (2017)). Banks with greater deposit market power tend to operate in more concentrated markets, offer lower deposit rates, and experience slower deposit growth. They also have a smaller proportion of time deposits, suggesting that their customers prioritize banking services over higher deposit rates.

Second, banks with greater deposit market power tend to have longer asset maturities and exhibit a greater degree of maturity mismatch. Both the maturity for loans and non-loan asset are higher for these banks, offering direct empirical support that they hold more long-term assets to stabilize earnings.[25]

Third, the asset composition differs between the two groups of banks. Specifically, banks with greater deposit market power hold more short-term liquid assets, such as cash and short-

[25] If difference in maturity patterns were driven solely by borrower characteristics, banks could simply hold shorter-term securities or securitize more long-term mortgages.

term securities, as well as more long-term securities and first-lien single-family mortgages.[26] Meanwhile, they hold fewer other real estates secured loans and Commercial and Industrial (C&I) Loans. There is no significant difference in the risk profiles of the loans between the two groups, as banks with greater deposit market power have a slightly higher non-performing loans ratio but a slightly lower net charge-off rate.

In principle, differences in asset composition could impact the results if they lead to differential changes in the interest rate sensitivity of lending profits between the two groups as the risk-free rate declines. In Section 4.4, I address this concern by directly evaluating changes in the interest rate sensitivity of lending profits, demonstrating that it does not differ significantly between the two groups when the risk-free rate decreases.

Verifying the Condition for Accurate Classification. As discussed in the theoretical section, a bank's deposit rate-setting decisions will depend only on the risk-free rate and its deposit market power if: (1) borrowing non-deposit funding involves minimal financial friction, or (2) the bank is a net investor in liquid assets, particularly short-term liquid assets. Under these conditions, I can accurately classify banks based on how fast they raise deposit rates in a rising interest rate environment.

First, U.S. banks can borrow advances from Federal Home Loan Banks (FHLBs) when they need non-deposit funding, which involves minimal financial friction.[27] The FHLB system is a government-sponsored entity (GSE) that issues GSE securities, which are implicitly guaranteed by the U.S. government. As a result, the debts of FHLB system enjoy more favorable terms than

[26] I define cash as the aggregate of "Noninterest-bearing balances and currency and coin", interest-bearing balances, Federal funds, and repurchase agreements (Repos). In the call report schedule RC-A, banks provide a breakdown of the first two items. The first term primarily consists of cash in transactions, which is minimal, while the second term includes balances due from other banks and the Federal Reserve. Short-term securities are those due within one year, while long-term securities have maturities exceeding one year. Both short-term and long-term securities are primarily government bonds and agency MBS.

[27] FHLB advances are over-collateralized loans from the FHLB system, which consists of 12 regional FHLBs and an Office of Finance that acts as the system's gateway to capital markets. The system is cooperatively owned by member financial institutions, primarily commercial banks and thrift institutions. Ashcraft, Bech, and Frame (2010) provides a detailed description of the FHLB systems. Originally, the FHLB advances were designed for mortgage related lending. However, prior research (e.g. Frame, Hancock, and Passmore (2012)) have shown that banks also use them to finance other loans, as money is fungible.

any AAA-rated private corporations. Because FHLB system provides advances to banks with minimal markup beyond its borrowing costs, the borrowing costs of FHLB advances generally remain within 20 basis points above the yield of Treasury securities with similar maturities, as shown in Figure 9. Besides, Figure 10 shows that deposits and FHLB advances constitute more than 99% of the median bank's total liabilities, implying banks primary rely on FHLB advances for non-deposit funding.[28] Therefore, banks' non-deposit borrowing experiences minimal financial friction, satisfying the first condition.

Meanwhile, US banks held significant amount of liquid asset during 2015-2018. As reflected in Table 3, the median ratio of liquid assets to total assets, including both short-term liquid assets and long-term securities, was 17.34% for low deposit market power banks and 25.54% for high deposit market power banks. Within this category, banks held large amounts of short-term liquid assets, primarily consisting of 'interest-paying' cash and short-term securities. The median ratio of short-term liquid assets was 7.54% for low deposit market power banks and 9.17% for high deposit market power banks. Additionally, the median ratio of first-lien single-family mortgages was 12.49% and 15.90% for low and high deposit market power banks, respectively. These mortgages can be securitized and sold if banks require funding, as prior literature has shown that there is little friction in securitizing these mortgages.

Because banks could access non-deposit funding sources with minimal friction and held ample liquid assets during 2015–2018, their deposit rate-setting decisions were primarily determined by their deposit market power and the risk-free rate. Consequently, classifying banks based on their deposit rate-setting decisions during 2015-2018 is justified. With these classifications, I proceed to examine how deposit betas change differentially for banks with high and low deposit market power as the economy enters low-interest-rate environments.

[28] Similar patterns also hold in the cross-section. As shown in Figure 11, for more than 91% of banks, deposits and FHLB advance make up more than 98% of their liabilities.

4.2 The Level of the Risk-Free Rate and the Differential Change in Banks' Deposit Betas

The theory suggests that deposit betas are lower in low-interest-rate environments, with a larger decline for banks with greater deposit market power. In this section, I empirically test these cross-sectional predictions, extending the analysis presented in Section 3.3.

Specifically, I first estimate Equation 28 for high and low deposit market power banks separately using data spanning from 2014Q2 to 2024Q2. As shown in column 1 and column 2 of Table 4, both groups of banks experience a decrease in deposit betas in low-interest-rate environments, but the deposit beta of banks with greater deposit market power decreases significantly more (0.173 vs. 0.117). These verify the theoretical prediction that the deposit beta decreases more for banks with greater deposit market power, when the economy enters low-interest-rate environments.

To formally test the hypothesis, I then estimate the following Equation,

$$\begin{aligned}\Delta r_{i,t}^{COF} = \alpha_i + \sum_{\tau=0}^{3} \xi_\tau^D \Delta r_{t-\tau} \mathbb{1}(r_{t-\tau} \leq \tilde{r}) \text{HighMP}_i + \sum_{\tau=0}^{3} \gamma_\tau^D \Delta r_{t-\tau} \mathbb{1}(r_{t-\tau} \leq \tilde{r}) \\ + \sum_{\tau=0}^{3} \theta_\tau^D \Delta r_{t-\tau} \text{HighMP}_i + \sum_{\tau=0}^{3} \beta_\tau^D \Delta r_{t-\tau} + \beta \text{Control}_{i,t} + \varepsilon_{i,t}\end{aligned} \tag{32}$$

The estimate of $\xi^D = \sum_{\tau=0}^{3} \xi_\tau^D$ captures the additional change in deposit betas from a high interest rate environment to a low interest rate environment when comparing high deposit market power banks to low deposit market power banks. The results are collected in Table 5.

As shown in column (2) of Table 5, the average deposit beta of banks with high deposit market power decreases by 0.065 more compared to banks with low deposit market power when transitioning to a low interest rate environment. In column (3) and column (4), I repeat the analysis using different thresholds (i.e., 1% and 2%) to define low interest rate environments. As shown in columns (3) and (4), the results remain consistent and the magnitudes for ξ^D are largely unchanged.

4.3 The Level of the Risk-Free Rate and Banks' Asset Maturity

In Sections 3.3 and 4.2, I found that deposit betas are smaller in low-interest-rate environments, with stronger decline for banks with greater deposit market power. This implies that the interest rate sensitivity of deposit profits is higher in low-interest-rate environments, with stronger effects for banks with higher deposit market power. The theory suggests that in such environments, all banks tend to increase their asset maturity, with those possessing greater deposit market power increasing their asset maturity more. In this section, I use the Fed funds rate cut induced by the Covid-19 pandemic to test these hypotheses by comparing the asset maturity of banks with high and low deposit market power before and after the rate cut.

Specifically, I conduct the following two diff-in-diff regressions.

$$q_{it} = \beta_1(\text{High MP}_i \times \text{Post}_t) + \beta_2 \text{High MP}_i + \beta_3 \text{Post}_t + \beta \text{Control}_i + \varepsilon_{it} \tag{33}$$

$$q_{it} = \beta_1(\text{High MP}_i \times \text{Post}_t) + \alpha_i + \delta_t + \varepsilon_{it} \tag{34}$$

Where q_{it} represents the asset maturity of bank i at date t, High MP_i is a binary variable indicating whether bank i belongs to the high deposit market power group. Post_t is a binary variable indicating whether date t is after 2020Q2. Control_i includes banks' Log asset value, tier1 capital ratio, deposit growth, and deposit ratio in 2019Q4. Additionally, α_i and δ_t denote bank fixed effects and time fixed effects, respectively. The sample period for these regressions are 2018Q4 to 2021Q4.[29] At last, the standard errors are clustered at the bank level for all diff-in-diff regressions.

As Table 6 shows, both groups of banks increased their asset maturity after the Fed funds rate dropped to zero. However, high deposit market power banks increase their asset maturity significantly more compared to low deposit market power banks. Specifically, following the

[29]Notice that there is no overlap between the classification period and the diff-in-diffs analysis period, even though both include 2018Q4. The classification uses the deposit rate in 2018Q4, which is calculated as the total deposit interest expense in 2018Q4 divided by the average deposits in 2018Q4, multiplied by four. In contrast, the diff-in-diffs analysis uses asset maturity at the end of 2018Q4.

drop in the Fed funds rate to zero after 2020Q2, low deposit market power banks increased their asset maturity by 0.416 years (4.99 months), while banks with higher deposit market power increased their asset maturity by an additional 0.134 years (1.61 months). The regression also confirms that high deposit market power banks have higher asset maturity.

Dynamic Effects: To understand how the shock took effects and test for pre-trend, I then conduct a dynamic diff-in-diff study, with results presented in Figure 12. As the figure shows, differences in asset maturity between high and low deposit market power banks remained relatively stable from 2018Q4 to 2019Q3, when the risk-free rates were stable. In 2019Q3, the Fed Reserve's rate cut led to a slight increase of this difference. In March 2020, Fed slashed Fed funds rate to zero in response to the Covid-19 pandemic. However, quarantines and other disruptions significantly impacted banking operations, causing asset maturity declines across all banks (as can be seen from Figure 3), with a slightly larger decrease for banks with greater deposit market power. After 2020Q2, as pandemic impacts eased and the economy shifted to a low interest rate environment, the difference in asset maturity between the two groups gradually increased.

In summary, Figure 12 suggests that banks with greater deposit market power increase their asset maturity more when the risk-free rate declines, likely to offset their larger increase in the interest rate sensitivity of the deposit profits under lower risk-free rates. These results thus provide further evidence that banks increase their asset maturity in response to the increased risks of their deposit profits.

4.4 The Interest Rate Sensitivity of Lending Profit

As discussed in my theoretical analysis, one implicit assumption is that the interest rate sensitivity of banks' lending profits does not change when the economy enters low-interest-rate environments. In the context of this diff-in-diff analysis, the assumption is that changes in interest rates do not differentially affect the interest rate sensitivity of lending profits between

banks with high and low deposit market power. A key concern, however, is that differences in asset composition across these groups might lead to differential changes in this interest rate sensitivity. If this occurs, the observed differential changes in asset maturity could be driven by the differential change in the interest rate sensitivity of lending profits rather than by the interest rate sensitivity of deposit profits alone.

To address these assumptions, I empirically demonstrate that the interest rate sensitivity of lending profits does not change when the economy enters low-interest-rate environments for either group of banks. Moreover, there is no differential change in this interest rate sensitivity between the two groups of banks. However, banks with greater deposit market power increase the maturity of their non-loan assets more, which suggests that these banks differentially increased their asset maturity in response to the differential change in the interest rate sensitivity of the deposit profits.

To assess the interest rate sensitivity of lending profits, I estimate the interest rate sensitivity of the lending income rate, denoted as lending income beta, by running regression 35 and use it to evaluate the interest rate sensitivity of lending profits. The lending income rate is defined as the difference between total interest income from loans and the corresponding net charge-offs, divided by total assets.

$$\Delta r_{it}^{LI} = \alpha_i + \sum_{\tau=0}^{3} \beta_\tau^{LI} \Delta r_{t-\tau} + \sum_{\tau=0}^{3} \gamma_\tau^{LI} \Delta r_{t-\tau} \mathbb{1}(r_{t-\tau} \leq \tilde{r}) + \varepsilon_{i,t} \tag{35}$$

As Table 7 shows, the interest rate sensitivity of the lending profits does not change significantly when the economy enters low-interest-rate environments for either group of banks. More importantly, it also shows that there is no differential change in this interest rate sensitivity between the two bank groups.[30] To rigorously test the hypothesis, I pool banks together and conduct the

[30]The insignificant negative estimates for the change in lending income beta is likely due to the fact that the loans maturity increased when the risk-free rates were lower.

combined regression 36.

$$\Delta r_{i,t}^{LI} = \alpha_i + \sum_{\tau=0}^{3} \xi_\tau^{LI} \Delta r_{t-\tau} \mathbb{1}(r_{t-\tau} \leq \tilde{r}) \text{HighMP}_i + \sum_{\tau=0}^{3} \gamma_\tau^{LI} \Delta r_{t-\tau} \mathbb{1}(r_{t-\tau} \leq \tilde{r}) + \sum_{\tau=0}^{3} \theta_\tau^{LI} \Delta r_{t-\tau} \text{HighMP}_i + \sum_{\tau=0}^{3} \beta_\tau^{LI} \Delta r_{t-\tau} + \varepsilon_{i,t} \tag{36}$$

Similar to the discussion about the deposit beta, there will be no differential change in the interest rate sensitivity of the lending profit between the two groups if $\xi^{LI} = \sum_{\tau=0}^{3} \xi_\tau^{LI}$ is not significantly different from zero.

As shown in the Table 8, there is no evidence for differential change in the interest rate sensitivity of the lending profit between two groups. As a result, the theory predicts that banks with greater deposit market power will increase the maturity of non-loans assets more to match the additional changes in the interest rate sensitivity of their deposit profit. To formally test this hypothesis, I define banks' non-loan asset maturity as:

$$q_{i,t}^{NL} = \sum_{j=1}^{N_2} \omega_{i,t}^{j} rm_{i,t}^{j} \tag{37}$$

Here, $q_{i,t}^{NL}$ denotes the non-loans asset maturity for bank i at date t, $\omega_{i,t}^{j}$ is the portfolio weight for asset class j relative to total asset and $rm_{i,t}^{j}$ is the corresponding average repricing maturity for asset class j. Importantly, this calculation excludes loans, so the maturity reflects the weighted average asset maturity, when assuming loan maturities are zero. This approach helps address the issue of differing loan-to-asset ratios between the two groups of banks.[31] I then estimate the diff-in-diff regression 33 and 34 using q_{it}^{NL} as the dependent variable.

The result in Table 9 confirmed the prediction that banks with high deposit market power in-

[31] All results remain consistent if I use the weighted average maturity for only non-loans asset, i.e.

$$q_{i,t}^{NL} = \sum_{j=1}^{N_2} \frac{A_{i,t}^{j}}{\sum_{j=1}^{N_2} A_{i,t}^{j}} rm_{i,t}^{j}$$

Here, N_2 includes all assets other than loans and $A_{i,t}^{j}$ refers to the asset value for asset class j at date t for bank i.

crease the maturity of their non-loans assets more to match the additional changes in the interest rate sensitivity of their deposit profit. As the table shows, both groups of banks increased their non-loans asset maturity after 2020Q2, but banks with higher deposit market power increased their non-loan asset maturity significantly more.

In summary, I find that the interest rate sensitivity of lending profits does not change differentially between the two groups when the economy enters low-interest-rates environments. However, the maturity of assets, excluding loans, increases more for banks with higher deposit market power. This evidence directly addresses the concern regarding the interest rate sensitivity of lending profits, suggesting that banks with higher deposit market power increase their asset maturity more to match the additional increase in the interest rate sensitivity of their deposit profits.

5 Additional Empirical Results

5.1 The Stability of Banks' Leverage

The theoretical analysis suggests that banks' leverage ratios could also affect their asset maturity. However, as illustrated in Figure 13, banks' leverage ratios—defined as liabilities over assets—have remained stable over the past two decades, with minimal cross-sectional variation. The left y-axis shows that the median leverage ratio in the banking sector has consistently ranged between 88.5% and 91.5%. Meanwhile, the right y-axis indicates that the standard deviation of the leverage ratio has been approximately 3% each quarter over the past two decades. This stability suggests that the primary reason for banks to increase their asset maturity in low-interest-rate environments is the increased interest rate sensitivity of their deposit profits.

5.2 The Interest Rate Risk of Net Non-Interest Income.

In addition to taking deposits, issuing loans, and investing in securities, US banks generate income from various other sources, such as investment banking, underwriting, and loan servicing. Moreover, the operation of banks incurs non-interest expenses, including employee salaries and branch maintenance costs. These non-interest income and non-interest expenses could also introduce interest rate risk to banks' cash flow and drive the variation in banks' asset maturity.

To rule out this concern, I estimate the interest rate risk of banks' non-interest income, non-interest expense, and the combined cash flow from these two sources by conducting the following analysis.

$$\Delta r_{it}^{j} = \alpha_i + \sum_{\tau=0}^{3} \beta_{\tau}^{j} \Delta r_{t-\tau} + \sum_{\tau=0}^{3} \gamma_{\tau}^{j} \Delta r_{t-\tau} \mathbb{1}(r_{t-\tau} \leq \bar{r}) + \beta \text{Control}_{i,t} + \varepsilon_{i,t} \tag{38}$$

Here, the rates include the Net Non-interest Income Rate (j = NNI), defined as the difference between non-interest income and non-interest expense divided by total assets; the Non-interest Income Rate (j = NI), defined as non-interest income divided by total assets; and the Non-interest Expense Rate (j = NE), defined as non-interest expense divided by total assets. Similar to deposit beta, a source of cash flow j will be subject to interest rate risk if the estimated $\beta^j = \sum_{\tau=0}^{3} \beta_{\tau}^{j}$ or $\gamma^j = \sum_{\tau=0}^{3} \gamma_{\tau}^{j}$ is significantly different from zero.

As shown in Table 10, neither non-interest income, non-interest expenses, nor their combined cash flow exhibit significant correlation with the Fed funds rate in either high or low interest rate environments. This lack of correlation suggests that these components of banks' cash flows are not exposed to interest rate risk.

6 Conclusion

This paper demonstrates that capital regulation can incentivize banks to maintain stable earnings, as failing to do so prevents banks from capturing profitable lending opportunities. Due to

banks' deposit market power, the pass-through from the risk-free rate to deposit rates is incomplete, which implies that banks' deposit profit increase with the risk-free rate. Because the profit from investing in long-term assets, net of funding costs, decreases as risk-free rates rise, holding long-term assets helps banks maintain stable earnings. In low-interest-rate environments, the pass-through from the risk-free rate to deposit rates becomes more sluggish (i.e. deposit betas are smaller), prompting banks to increase their asset maturity.

In the empirical analysis, I confirm that deposit betas are smaller in low-interest-rate environments. Additionally, I find that these deposit betas decline more for banks with greater deposit market power. Using the exogenous Fed funds rate reduction induced by the COVID-19 pandemic, I further find that asset maturity increases more for banks with greater deposit market power when the Fed funds rate decreases. I also confirmed previous empirical results that banks' earnings have little correlation with the risk-free rate and showed that this results hold in both high- and low-interest-rate environments.

Overall, this paper analyzes banks' risk management strategies and shows that these strategies compel banks to maintain stable earnings. Due to the nature of banks' deposit business, their securities portfolios become important tools for achieving this risk management goal. As a result, asset maturities vary across different interest rate environments. By linking banks' asset maturity choices and interest rate risk to capital regulation, this study opens new avenues for exploring optimal regulatory frameworks that address both interest rate risk and credit risk. Additionally, given banks' pivotal role in the fixed-income market, a low risk-free rate policy that induces banks to invest more in long-term assets may contribute to a reduction in the term premium. Consequently, the mechanisms uncovered in this paper have important implications for the study of both optimal bank regulation and monetary policy transmission.

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Table 1: Deposit Beta in (Relatively) High and Low Interest Rate Environments. This table summarizes the results from running the following regression:

$$\Delta r_{i,t}^{COF} = \alpha_i + \sum_{\tau=0}^{3} \beta_\tau^D \Delta r_{t-\tau} + \sum_{\tau=0}^{3} \gamma_\tau^D \Delta r_{t-\tau} \mathbb{1}(r_{t-\tau} \leq \tilde{r}) + \beta \text{Control}_{i,t} + \varepsilon_{i,t}$$

$\beta^D = \sum_{\tau=0}^{3} \beta_\tau^D$ represents the estimated deposit beta when the risk-free rates are relatively high. Similarly, $\gamma^D = \sum_{\tau=0}^{3} \gamma_\tau^D$ denotes the estimated change in deposit beta in low-interest-rate environments compared to the period when the risk-free rates are relatively high. $\mathbb{1}(r_t \leq \tilde{r})$ equals 1 if r_t is less than the threshold $\tilde{r}$, and 0 otherwise. Controls include the bank's log of asset value and Tier-1 capital ratio. In columns (1), I estimate the deposit beta without differentiating between various interest rate environments, which is the same as setting the threshold below zero. In column (2), (3), and (4), the threshold $\tilde{r}$ equals 1.5%. In columns (5) and column (6), the threshold $\tilde{r}$ equals 1% and 2%, respectively. The sample period for Column (1), (2), (4), (5), and (6) is 2002Q1-2024Q2, and the sample period for column (3) is 2014Q2-2024Q2. In column (4), the sample is restricted to the 200 largest banks as of 2018Q4, whereas all other columns use the full sample. Standard errors are double clustered at bank and time level and are collected in parentheses. * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

	(1)	(2)	(3)	(4)	(5)	(6)
$\tilde{r}$	NA	1.5%	1.5%	1.5%	1%	2%
β^D	0.331^{***}	0.370^{***}	0.373^{***}	0.468^{***}	0.381^{***}	0.372^{***}
	(0.026)	(0.022)	(0.015)	(0.026)	(0.020)	(0.024)
γ^D		-0.141^{***}	-0.152^{***}	-0.161^{***}	-0.224^{***}	-0.120^{***}
		(0.052)	(0.032)	(0.061)	(0.054)	(0.044)
Control	Yes	Yes	Yes	Yes	Yes	Yes
Bank FE	Yes	Yes	Yes	Yes	Yes	Yes
R^2	0.4639	0.4747	0.5061	0.3930	0.4844	0.4752
Num of Obs	271,363	271,363	140,072	14,318	271,363	271,363

Table 2: The Interest Rate Risk of Banks' Earnings. This table summarizes the results from running the following regression:

$$\Delta\text{ROA}_{i,t} = \alpha_i + \sum_{\tau=0}^{3} \beta_\tau^{ROA} \Delta r_{t-\tau} + \sum_{\tau=0}^{3} \gamma_\tau^{ROA} \Delta r_{t-\tau} \mathbb{1}(r_{t-\tau} \leq \tilde{r}) + \beta \text{Control}_{i,t} + \varepsilon_{i,t}$$

$\beta^{ROA} = \sum_{\tau=0}^{3} \beta_\tau^{ROA}$ represents the estimated ROA beta when the risk-free rates are relatively high. Similarly, $\gamma^{ROA} = \sum_{\tau=0}^{3} \gamma_\tau^{ROA}$ denotes the estimated change in beta in low-interest-rate environments compare to the period when the risk-free rates are relatively high. $\mathbb{1}(r_{t-\tau} \leq \tilde{r})$ equals 1 if r_t is less than the threshold $\tilde{r}$, and 0 otherwise. The specific thresholds $\tilde{r}$ used is listed in the table. In the first column, I don't include the interaction terms, which is the same as having threshold $\tilde{r} = 0$. Controls include the banks' log of asset value and Tier-1 capital ratio. The sample period is 2002Q1-2024Q2. Standard errors are double clustered at bank and time level and are collected in parentheses. $^{*}\, p < 0.1$; $^{**}\, p < 0.05$; $^{***}\, p < 0.01$.

	(1)	(2)	(3)	(4)
$\tilde{r}$	NA	1.5%	1%	2%
β^{ROA}	0.029	0.021	0.053	0.004
	(0.042)	(0.047)	(0.047)	(0.050)
γ^{ROA}		0.006	−0.097	0.056
		(0.162)	(0.160)	(0.144)
Control	Yes	Yes	Yes	Yes
Bank FE	Yes	Yes	Yes	Yes
R^2	-0.0066	-0.0056	-0.0060	-0.0054
Num of Obs	271,363	271,363	271,363	271,363

Table 3: Summary Statistics for High and Low Deposit Market Power Banks This table summarizes the relevant bank characteristics by market power groups. All variables are calculated based on banks' financial statements from 2018Q4, except for the deposit market HHI, which is based on the FDIC Summary of Deposits from 2018Q2, as the FDIC collects this information only in the second quarter of each year. The maturity of non-loan assets is calculated using Equation 37, which calculates asset maturity under the assumption that loans have zero maturity. Section 4.4 provides a detailed discussion and rationale for this approach. The calculation of deposit market HHI follows Drechsler, Savov, and Schnabl (2021). 'SFR Mortgage' refers to first-lien single-family residential mortgage. The net charge-off rate is defined as the difference between charge-offs and recoveries, divided by the quarterly average loan amount. 'Non-performing Loans' refers to loans classified as either 'Past due 90 days or more and still accruing' or 'Nonaccrual.' All variables are winsorized at the 95% level.

	High Deposit M.P.		Low Deposit M.P.	
Variables	**Mean**	**Median**	**Mean**	**Median**
Log(Assets)	8.55	8.45	8.81	8.68
Asset Maturity (years)	5.39	4.94	4.71	4.30
Loans Maturity (years)	4.27	3.75	3.79	3.33
Non-loans Asset Maturity (years)	3.36	2.84	2.44	1.95
Deposit Ratio (%)	95.22	97.03	92.49	93.68
Time Deposit Ratio (%)	25.88	24.62	36.56	35.23
Deposit Rate (%)	0.58	0.57	1.14	1.09
Deposit Growth (%)	3.66	2.35	8.65	7.02
Deposit Market HHI	1646.26	1418.69	1380.57	1171.43
Short-Term Liquid Asset Ratio (%)	11.14	9.17	9.06	7.54
Long Term Securities Ratio (%)	18.25	16.37	12.06	9.80
SFR Mortgage Ratio (%)	18.65	15.90	15.42	12.49
Other Real Estate Loans Ratio (%)	31.84	31.91	37.56	38.45
C&I Loans Ratio (%)	7.51	6.50	10.35	9.01
Consumer Loans Ratio (%)	2.79	1.79	2.66	1.22
Other Loans Ratio (%)	4.07	1.76	5.31	1.98
Net Charge-Off Rate (%)	0.15	0.03	0.21	0.04
Non-performing Loans Ratio (%)	0.88	0.57	0.77	0.47
ROA (%)	1.11	1.09	1.13	1.14
Tier1-Capital Ratio (%)	17.28	15.50	14.46	12.98
Number of Observations	2399	2399	1173	1173

Table 4: Deposit Beta in (Relatively) High and Low Interest Rate Environments for High and Low Deposit Market Power Banks. This table summarizes the results from running the following regression separately for high and low deposit market power banks:

$$\Delta r_{i,t}^{COF} = \alpha_i + \sum_{\tau=0}^{3} \beta_\tau^D \Delta r_{t-\tau} + \sum_{\tau=0}^{3} \gamma_\tau^D \Delta r_{t-\tau} \mathbb{1}(r_{t-\tau} \leq \tilde{r}) + \beta \text{Control}_{i,t} + \varepsilon_{i,t}$$

$\beta^D = \sum_{\tau=0}^{3} \beta_\tau^D$ represents the estimated deposit beta when the risk-free rates are relatively high. Similarly, $\gamma^D = \sum_{\tau=0}^{3} \gamma_\tau^D$ denotes the estimated change in deposit beta in low-interest-rate environments compared to the period when the risk-free rates are relatively high. In column (1) and (2), the threshold $\tilde{r}$ is 1.5%. In columns (3) and (4), the threshold $\tilde{r}$ is 1%. The sample period is 2014Q2-2024Q2. Standard errors are double-clustered at the bank and time levels and are reported in parentheses. $^{*}\, p < 0.1$; $^{**}\, p < 0.05$; $^{***}\, p < 0.01$.

	(1)	(2)	(3)	(4)	(5)	(6)
$\tilde{r}$	1.5%	1.5%	1%	1%	2%	2%
Market Power	High	Low	High	Low	High	Low
β^D	0.325^{***}	0.454^{***}	0.325^{***}	0.460^{***}	0.325^{***}	0.453^{***}
	(0.017)	(0.016)	(0.015)	(0.014)	(0.018)	(0.018)
γ^D	-0.173^{***}	-0.117^{***}	-0.223^{***}	-0.178^{***}	-0.135^{***}	-0.090^{**}
	(0.032)	(0.037)	(0.035)	(0.037)	(0.032)	(0.038)
Control	Yes	Yes	Yes	Yes	Yes	Yes
Bank FE	Yes	Yes	Yes	Yes	Yes	Yes
R^2	0.4870	0.5542	0.4923	0.5581	0.4858	0.5532
Num of Obs	86,901	44,481	86,901	44,481	86,901	44,481

Table 5: Deposit Beta in (Relatively) High and Low Interest Rate Environments for High and Low Deposit Market Power Banks in A Combined Regression. This table summarizes the results from running the following regression:

$$\Delta r_{i,t}^{COF} = \alpha_i + \sum_{\tau=0}^{3} \xi_\tau^D \Delta r_{t-\tau} \mathbb{1}(r_{t-\tau} \leq \tilde{r}) \text{HighMP}_i + \sum_{\tau=0}^{3} \gamma_\tau^D \Delta r_{t-\tau} \mathbb{1}(r_{t-\tau} \leq \tilde{r})$$
$$+ \sum_{\tau=0}^{3} \theta_\tau^D \Delta r_{t-\tau} \text{HighMP}_i + \sum_{\tau=0}^{3} \beta_\tau^D \Delta r_{t-\tau} + \beta \text{Control}_{i,t} + \varepsilon_{i,t}$$

$\beta^D = \sum_{\tau=0}^{3} \beta_\tau^D$ represents the deposit beta for banks with low deposit market power when the risk-free rates are relatively high. $\theta^D = \sum_{\tau=0}^{3} \theta_\tau^D$ signifies the difference in deposit beta between high deposit market power banks and low deposit market power banks when the risk-free rate is relatively high. $\gamma^D = \sum_{\tau=0}^{3} \gamma_\tau^D$ indicates the change in deposit beta when comparing low interest rate environments to period when the risk-free rates are relatively high for low deposit market power banks. $\xi^D = \sum_{\tau=0}^{3} \xi_\tau^D$ denotes the additional change in deposit beta for high deposit market power banks compared to low deposit market power banks when comparing low interest rate environments to period when the risk-free rates are relatively high. $\mathbb{1}(r_{t-\tau} \leq \tilde{r})$ equals 1 if r_t is less than $\tilde{r}$, and 0 otherwise. The sample period is 2014Q2-2024Q2. For columns (1) and (2), the threshold $\tilde{r}$ is 1.5%. For columns (3) and (4), the thresholds $\tilde{r}$ are 1% and 2% respectively. Standard errors are double-clustered at the bank and time levels and are reported in parentheses. $^{*}\, p < 0.1$; $^{**}\, p < 0.05$; $^{***}\, p < 0.01$.

	(1)	(2)	(3)	(4)
$\tilde{r}$	1.5%	1.5%	1%	2%
β^D	0.461***	0.451***	0.456***	0.449***
	(0.013)	(0.016)	(0.014)	(0.018)
γ^D	−0.132***	−0.112***	−0.171***	−0.085**
	(0.035)	(0.037)	(0.038)	(0.038)
θ^D	−0.129***	−0.124***	−0.129***	−0.122***
	(0.011)	(0.013)	(0.012)	(0.013)
ξ^D	−0.061***	−0.065***	−0.057**	−0.054**
	(0.021)	(0.022)	(0.027)	(0.021)
Control	No	Yes	Yes	Yes
Bank FE	Yes	Yes	Yes	Yes
R^2	0.5437	0.5212	0.5258	0.5202
Num of Obs	150,079	131,382	131,382	131,382

Table 6: Diff-in-Diff Analysis on Asset Maturity. This table summarizes the results from running the following regressions:

$$q_{it} = \beta_1(\text{High MP}_i \times \text{Post}_t) + \beta_2\text{High MP}_i + \beta_3\text{Post}_t + \beta\text{Control}_i + \varepsilon_{it}$$

$$q_{it} = \beta_1(\text{High MP}_i \times \text{Post}_t) + \alpha_i + \delta_t + \varepsilon_{it}$$

q_{it} is bank i's asset maturity at date t. High MP_i is a binary variable that indicates whether bank i belongs to the high deposit market power group. Post_t is a binary indicator that denotes whether the date t is post 2020Q2. The classification of the deposit market power group is based on the mixture method. In columns (1), (2), and (4), the sample period spans from 2018Q4 to 2021Q4. For results in columns (3), I exclude 2020Q2, to avoid the initial impact of the COVID-19 pandemic shock. The standard errors are clustered at the bank level and are included in the parentheses. $^{*}\,p < 0.1$; $^{**}\,p < 0.05$; $^{***}\,p < 0.01$.

	(1)	(2)	(3)	(4)
HighMP × Post	0.156^{***}	0.134^{***}	0.131^{***}	0.109^{***}
	(0.041)	(0.039)	(0.042)	(0.035)
Post	0.408^{***}	0.416^{***}	0.418^{***}	
	(0.032)	(0.031)	(0.033)	
HighMP	0.706^{***}	0.801^{***}	0.805^{***}	
	(0.089)	(0.092)	(0.092)	
Controls	No	Yes	Yes	No
Bank FE	No	No	No	Yes
Time FE	No	No	No	Yes
R^2	0.0308	0.1048	0.1043	0.9311
Num of Obs	40,723	39,932	36,840	40,671

Table 7: Lending Income Beta in (Relatively) High and Low Interest Rate Environments for Banks with High and Low Deposit Market Power. This table summarizes the results from running the following regression separately for high and low deposit market power banks:

$$\Delta r_{it}^{LI} = \alpha_i + \sum_{\tau=0}^{3} \beta_\tau^{LI} \Delta r_{t-\tau} + \sum_{\tau=0}^{3} \gamma_\tau^{LI} \Delta r_{t-\tau} \mathbb{1}(r_{t-\tau} \leq \tilde{r}) + \varepsilon_{i,t}$$

$\beta^{LI} = \sum_{\tau=0}^{3} \beta_\tau^{LI}$ represents the estimated lending income beta when the risk-free rates are relatively high. Similarly, $\gamma^{LI} = \sum_{\tau=0}^{3} \gamma_\tau^{LI}$ denotes the estimated change in this beta in low-interest-rate environments compared to the period when the risk-free rates are relatively high. In column (1) and (2), the threshold $\tilde{r}$ is 1.5%. In columns (3) and (4), the threshold $\tilde{r}$ is 1%. In columns (5) and (6), the threshold $\tilde{r}$ is 2%. The sample period is 2014Q2-2024Q2. Standard errors are double-clustered at the bank and time levels and are reported in parentheses. $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.

	(1)	(2)	(3)	(4)	(5)	(6)
$\tilde{r}$	1.5%	1.5%	1%	1%	2%	2%
Market Power	High	Low	High	Low	High	Low
β^{LI}	0.198^{***}	0.263^{***}	0.198^{***}	0.271^{***}	0.177^{***}	0.247^{***}
	(0.032)	(0.033)	(0.032)	(0.030)	(0.030)	(0.031)
γ^{LI}	-0.111	-0.090	-0.138	-0.141	-0.043	-0.042
	(0.074)	(0.087)	(0.082)	(0.089)	(0.063)	(0.071)
Control	Yes	Yes	Yes	Yes	Yes	Yes
Bank FE	Yes	Yes	Yes	Yes	Yes	Yes
R^2	-0.0115	0.0034	-0.0110	0.0045	-0.0129	0.0028
Num of Obs	76,705	38,227	76,705	38,227	76,705	38,227

Table 8: Lending Income Beta in (Relatively) High and Low Interest Rate Environments for High and Low Deposit Market Power Banks in A Combined Regression. This table summarizes the results from running the following regression:

$$\Delta r_{i,t}^{LI} = \alpha_i + \sum_{\tau=0}^{3} \xi_\tau^{LI} \Delta r_{t-\tau} \mathbb{1}(r_{t-\tau} \leq \tilde{r}) \text{HighMP}_i + \sum_{\tau=0}^{3} \gamma_\tau^{LI} \Delta r_{t-\tau} \mathbb{1}(r_{t-\tau} \leq \tilde{r})$$
$$+ \sum_{\tau=0}^{3} \theta_\tau^{LI} \Delta r_{t-\tau} \text{HighMP}_i + \sum_{\tau=0}^{3} \beta_\tau^{LI} \Delta r_{t-\tau} + \beta \text{Control}_{i,t} + \varepsilon_{i,t}$$

$\beta^{LI} = \sum_{\tau=0}^{3} \beta_\tau^{LI}$ represents the lending income beta for banks with low deposit market power when the risk-free rates are relatively high. $\theta^{LI} = \sum_{\tau=0}^{3} \theta_\tau^{LI}$ signifies the difference in lending income beta between high deposit market power banks and low deposit market power banks when the risk-free rate is relatively high. $\gamma^{LI} = \sum_{\tau=0}^{3} \gamma_\tau^{LI}$ indicates the change in lending income beta when comparing low interest rate environments to period when the risk-free rates are relatively high for low deposit market power banks. $\xi^{LI} = \sum_{\tau=0}^{3} \xi_\tau^{LI}$ denotes the additional change in lending income beta for high deposit market power banks compared to low deposit market power banks when comparing low interest rate environments to period when the risk-free rates are relatively high. $\mathbb{1}(r_{t-\tau} \leq \tilde{r})$ equals 1 if r_t is less than $\tilde{r}$, and 0 otherwise. The sample period is 2014Q2-2024Q2. The thresholds for columns (1), (2) and (3) are 1.5%, 1%, and 2%. Standard errors are double-clustered at the bank and time levels and are reported in parentheses. $^{*}\, p < 0.1$; $^{**}\, p < 0.05$; $^{***}\, p < 0.01$.

	(1)	(2)	(3)
$\tilde{r}$	1.5%	1%	2%
β^{LI}	0.273***	0.279***	0.256***
	(0.035)	(0.032)	(0.033)
γ^{LI}	−0.111	−0.161*	−0.058
	(0.090)	(0.094)	(0.073)
θ^{LI}	−0.082***	−0.086***	−0.085***
	(0.013)	(0.014)	(0.010)
ξ^{LI}	0.014	0.036	0.026
	(0.029)	(0.033)	(0.019)
Control	Yes	Yes	Yes
Bank FE	Yes	Yes	Yes
R^2	-0.0059	-0.0052	-0.0069
Num of Obs	114,932	114,932	114,932

Table 9: Diff-in-Diff Analysis on Non-Loans Asset Maturity. This table summarizes the results from running the following regressions:

$$q_{it}^{NL} = \beta_1(\text{High MP}_i \times \text{Post}_t) + \beta_2 \text{High MP}_i + \beta_3 \text{Post}_t + \text{Control}_i + \varepsilon_{it}$$

$$q_{it}^{NL} = \beta_1(\text{High MP}_i \times \text{Post}_t) + \alpha_i + \delta_t + \varepsilon_{it}$$

q_{it}^{NL} is bank i's non-loans asset maturity at date t, i.e. $q_{i,t}^{NL} = \sum_{j=1}^{N_2} \omega_{A,i,t}^{j} rm_{A,i,t}^{j}$ (see Section 4.4 for detailed definitions). High MP_i is a binary variable that indicates whether bank i belongs to the high deposit market power group. Post_t is a binary indicator that denotes whether the date t is post 2020Q2. The classification of the deposit market power group is based on the mixture model. The sample period is from 2018Q4 to 2021Q4. The standard errors are clustered at the bank level and are included in the parentheses. * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

	(1)	(2)	(3)
HighMP × Post	0.227***	0.217***	0.204***
	(0.036)	(0.034)	(0.030)
Post	0.270***	0.264***	
	(0.027)	(0.026)	
HighMP	0.959***	0.817***	
	(0.078)	(0.077)	
Controls	No	Yes	No
Bank FE	No	No	Yes
Time FE	No	No	Yes
R^2	0.0541	0.2126	0.9395
Num of Obs	40,723	39,932	40,671

Table 10: The Interest Rate Risk of Net Non-interest Income. This table summarizes the results from running the following regression:

$$\Delta r_{it}^{j} = \alpha_i + \sum_{\tau=0}^{3} \beta_{\tau}^{j} \Delta r_{t-\tau} + \sum_{\tau=0}^{3} \gamma_{\tau}^{j} \Delta r_{t-\tau} \mathbb{1}(r_{t-\tau} \leq \tilde{r}) + \beta \text{Control}_{i,t} + \varepsilon_{i,t}$$

$\beta^j = \sum_{\tau=0}^{3} \beta_{\tau}^{j}$ represents the estimated beta when the risk-free rates are relatively high for cash flow source j. Similarly, $\gamma^j = \sum_{\tau=0}^{3} \gamma_{\tau}^{j}$ denotes the estimated change in beta for cash flow source j in low-interest-rate environments compare to the period when the risk-free rates are relatively high. $\mathbb{1}(r_{t-\tau} \leq \tilde{r})$ equals 1 if r_t is less than the threshold 1.5%, and 0 otherwise. Controls include the bank's log of asset value and Tier-1 capital ratio. The sample period is 2002Q1-2024Q2. Standard errors are double clustered at bank and time level and are collected in parentheses. $^{*}\, p < 0.1$; $^{**}\, p < 0.05$; $^{***}\, p < 0.01$.

	(1)	(2)	(3)	(4)	(5)	(6)
j	NNI	NNI	NI	NI	NE	NE
β^j	-0.032	-0.008	0.003	0.009	0.036	0.017
	(0.034)	(0.031)	(0.012)	(0.013)	(0.028)	(0.036)
γ^j		-0.099		-0.048		0.058
		(0.097)		(0.033)		(0.086)
Control	Yes	Yes	Yes	Yes	Yes	Yes
Bank FE	Yes	Yes	Yes	Yes	Yes	Yes
R^2	-0.0082	-0.0070	-0.0132	-0.0122	-0.0055	-0.0046
Num of Obs	271,363	271,363	271,363	271,363	271,363	271,363

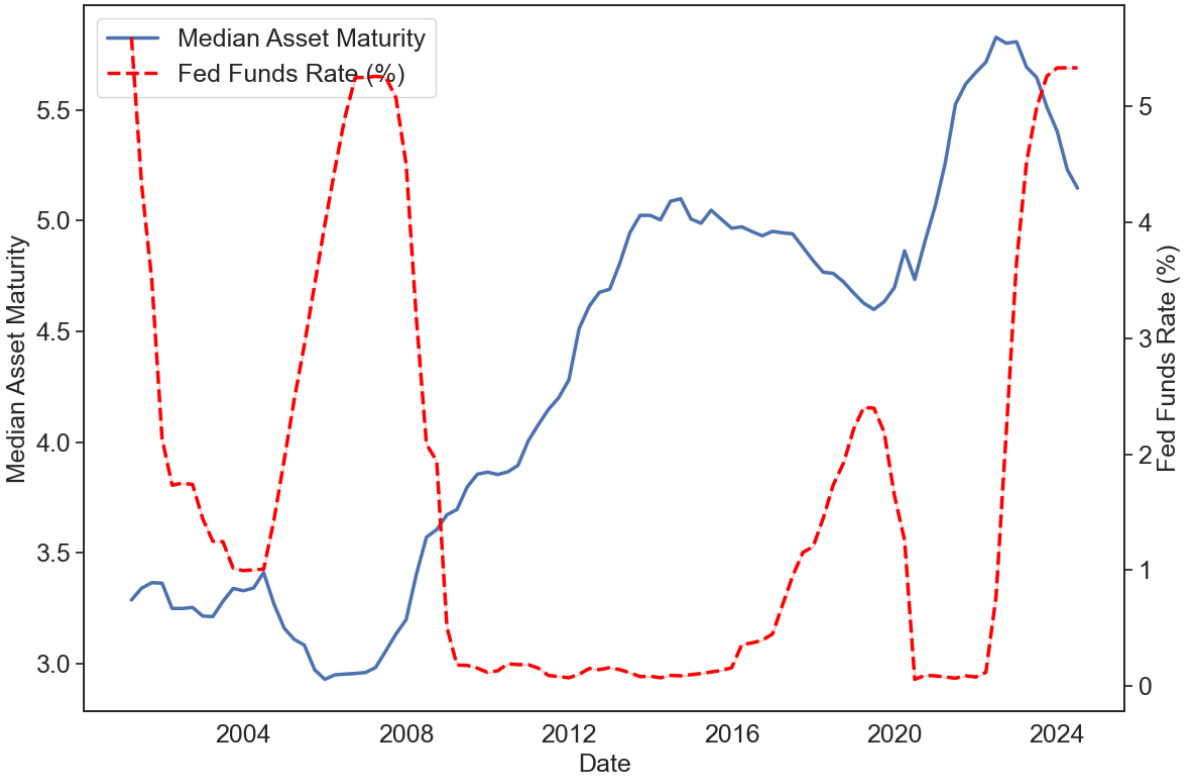

Figure 1: The Median Asset Maturity of the Banking Industry Over Time. This graph illustrates the median level of banks' asset maturity alongside the Fed funds rate. Asset maturity is calculated as the weighted average repricing maturity of banks.

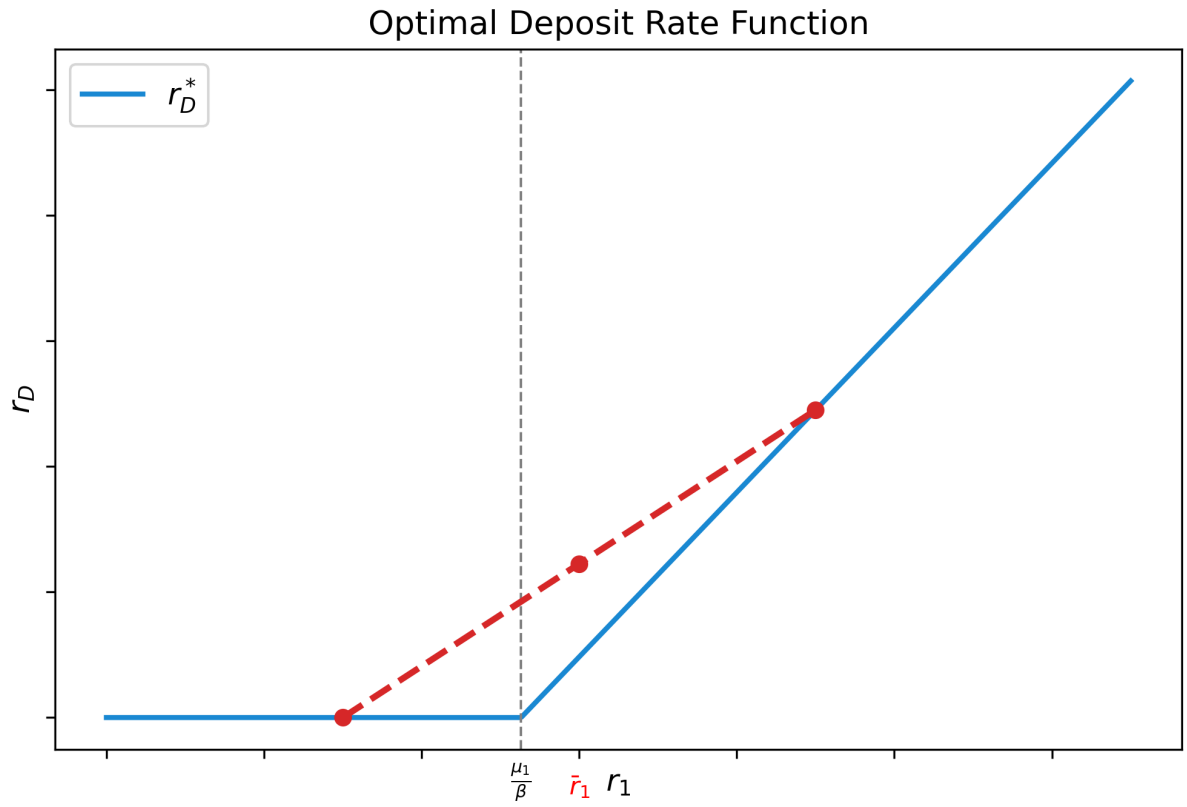

Figure 2: Banks' Optimal Deposit Rate Function in the Model. This graph illustrates the relationship between the optimal deposit rate and the risk-free rate. The blue line represents the optimal deposit rate function. The slope of the dashed red line corresponds to the deposit beta of this bank when the expected risk-free rate is $\bar{r}_1$.

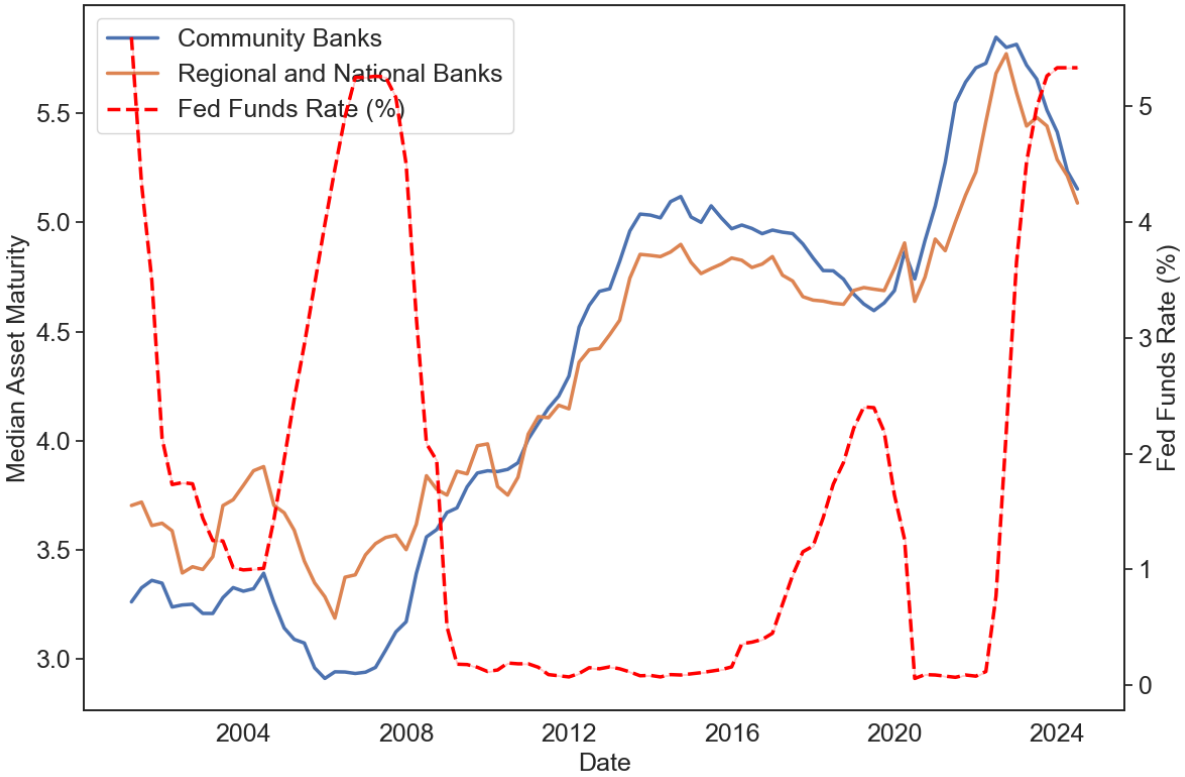

Figure 3: Median Asset Maturity for Community Banks vs. Regional and National Banks. This graph illustrates the median asset maturity levels of community banks vs. regional and national banks, alongside the Fed funds rate. Regional and national banks are defined as the 200 largest banks by asset size as of 2018Q4, while community banks include all remaining banks in the sample.

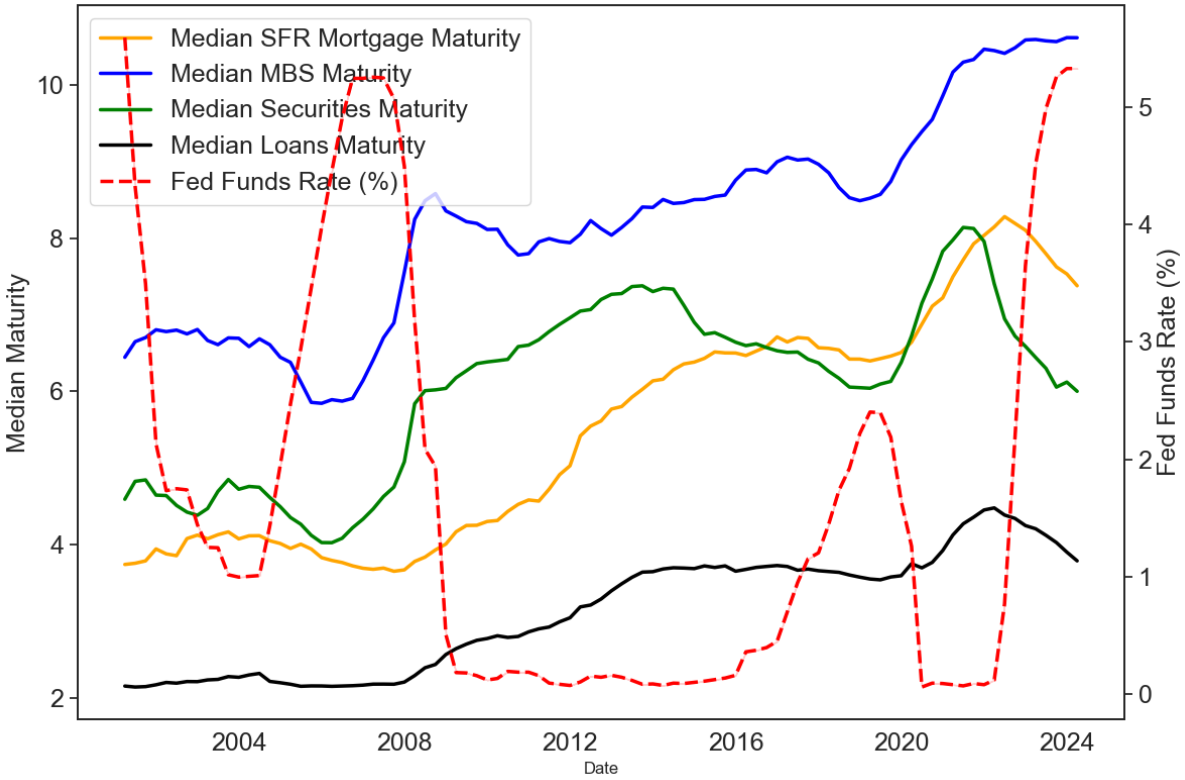

Figure 4: Median Maturity for Different Asset Classes. This graph illustrates the median maturity for banks' securities, single family residential mortgage, and loans, alongside with the Fed funds rate.

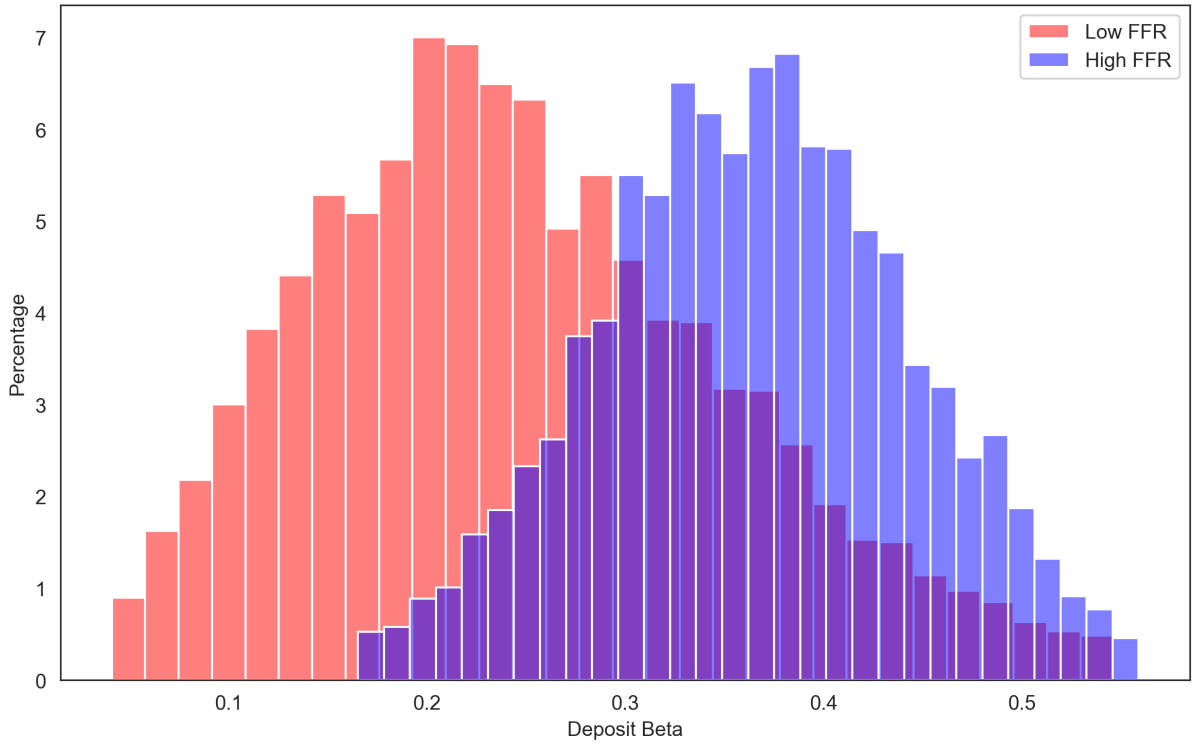

Figure 5: Deposit Betas in High and low Interest Rate Environment. In this figure, I plot the distribution of estimated deposit beta for 1996-2008 and 2009-2021, where the former represents the period where the Fed Funds Rate (FFR) is high and the later represents the period where FRR is low. The estimation of deposit betas follows Drechsler, Savov, and Schnabl (2021).

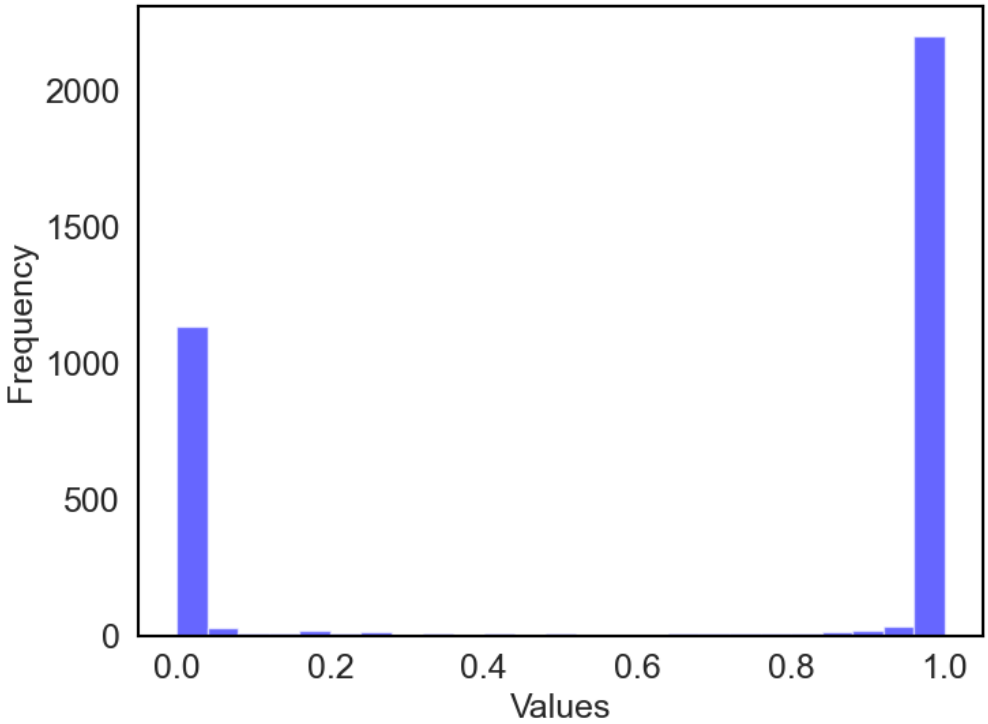

Figure 6: Posterior Probability for Banks to be in High Deposit Market Power Group. In this figure, I plot the posterior probability distribution for banks to be in high market power group.

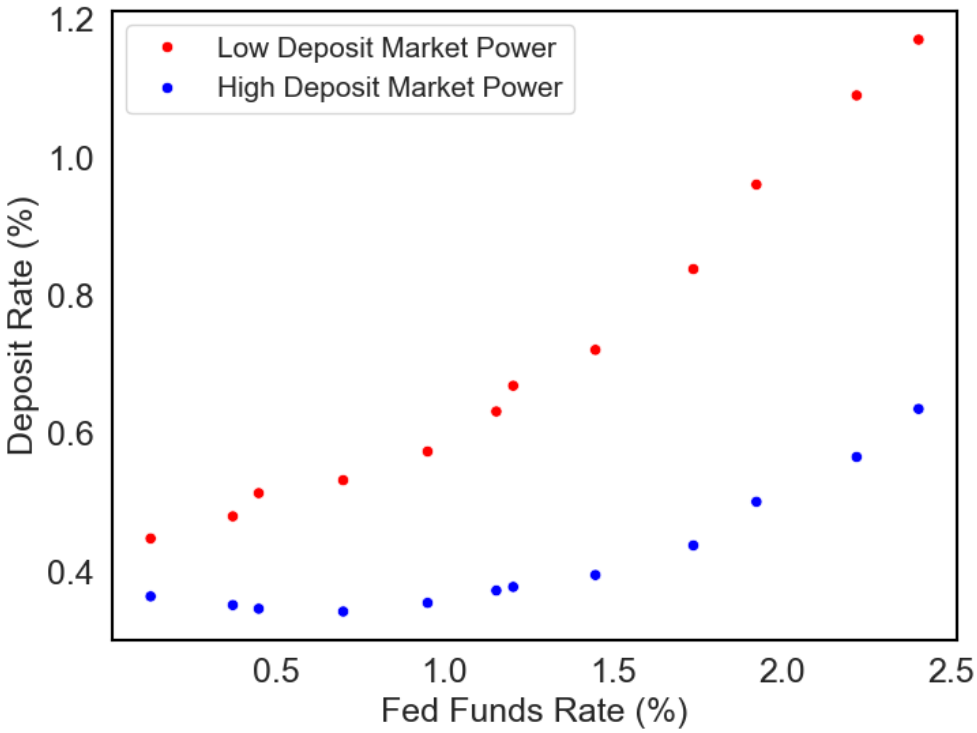

Figure 7: Deposit Rate Path for High and Low Deposit Market Power Banks During 2015-2018 Monetary Policy Tightening Period. In this figure, I plot the binscatter figure of deposit rates against Fed Funds rates for both groups of banks for 2015-2018 monetary policy tightening period. The group membership is based on the posterior probability distribution estimated using the mixture model.

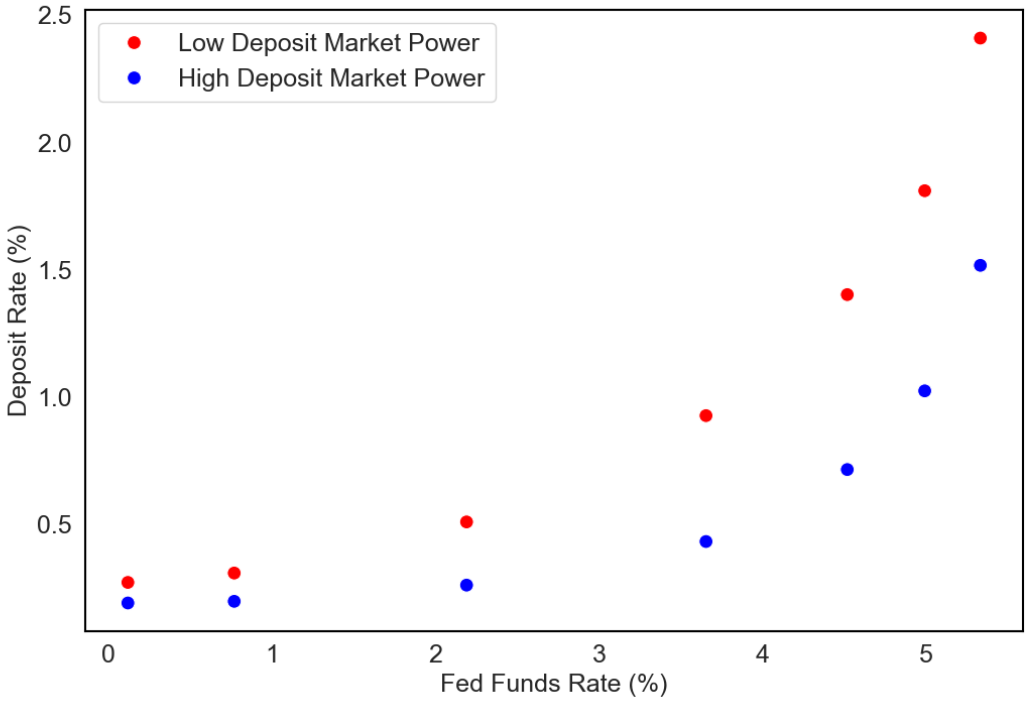

Figure 8: Deposit Rate Path for High and Low Deposit Market Power Banks During 2022-2024 Monetary Policy Tightening Period. In this figure, I plot the binscatter figure of deposit rates against Fed funds rate for each group for 2022-2024 monetary policy tightening period. Here, the group membership is based on the posterior probability distribution estimated using the mixture model.

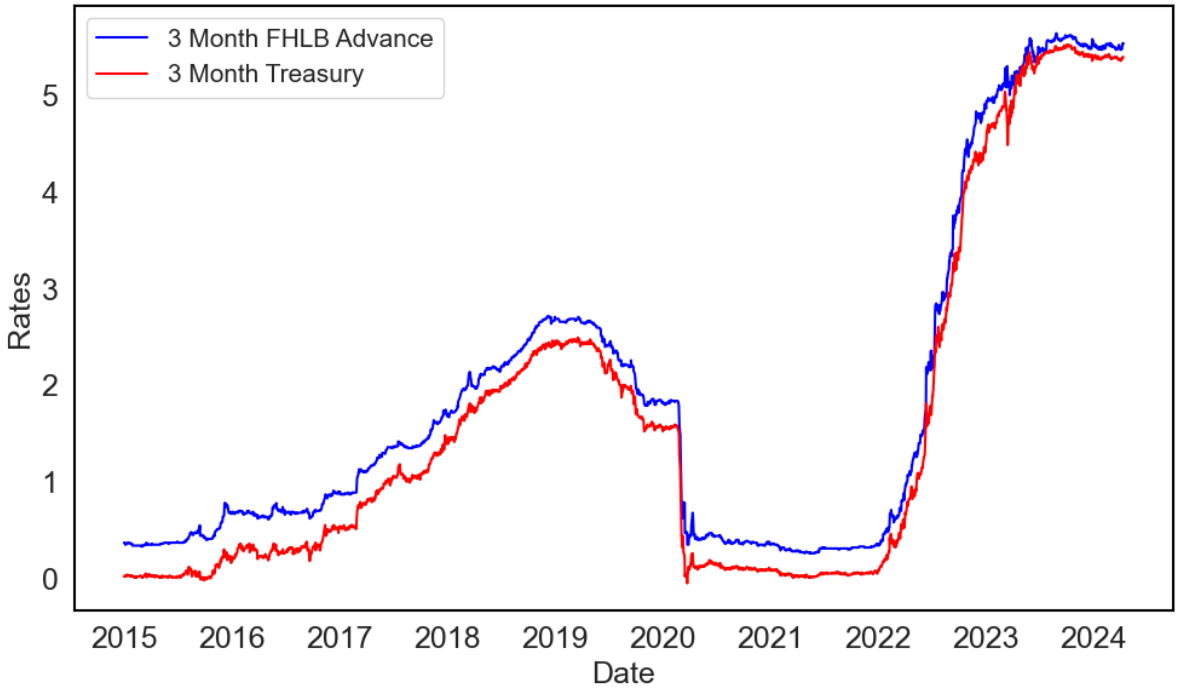

Figure 9: Comparison of Rates Between Treasury and FHLB Advance. In this figure, I plot the rates of 3 month Treasury bill and 3 month FHLB advances overtime.

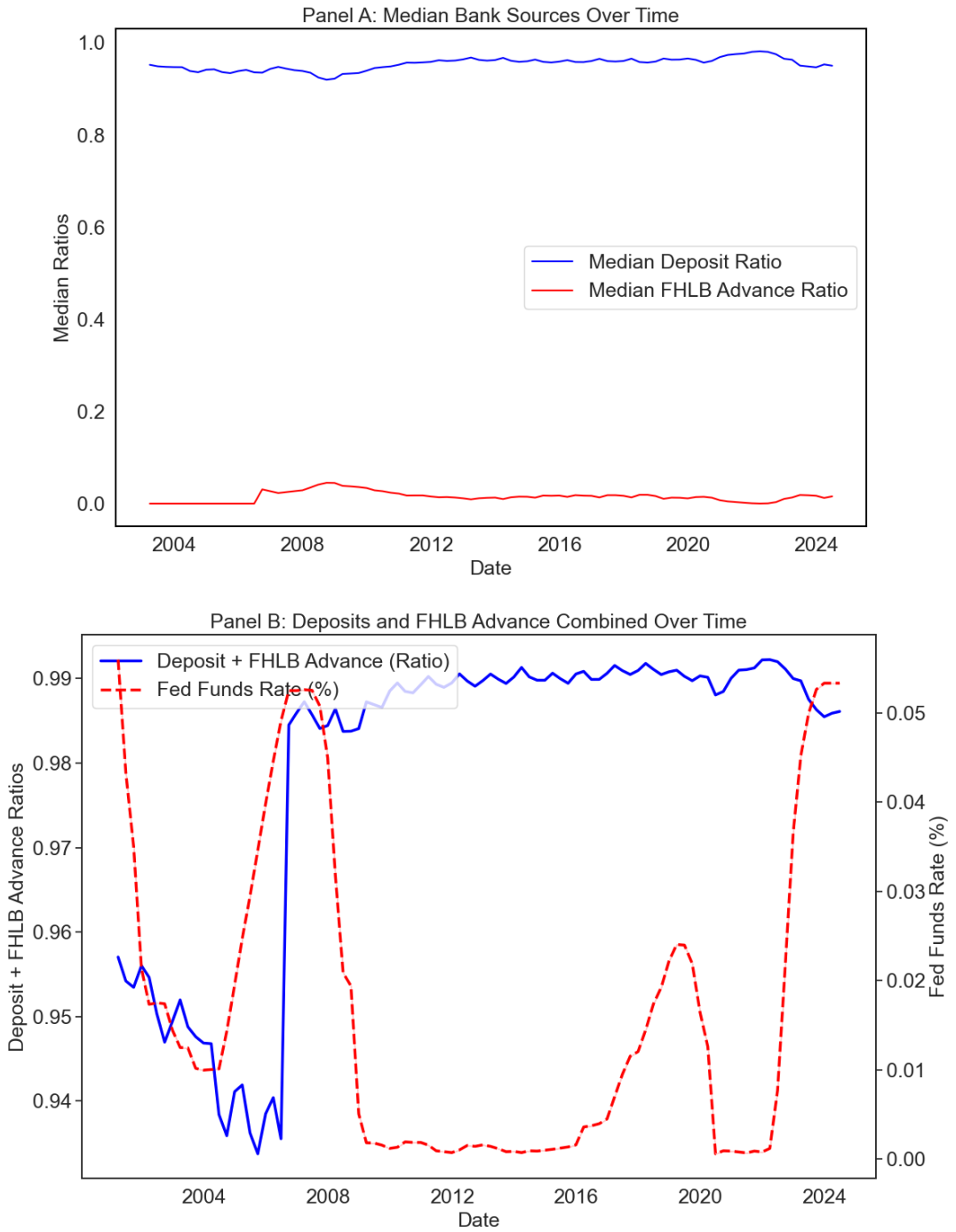

Figure 10: Decomposition of Banks Funding Sources. This figure provides an overview of the funding sources utilized by U.S. banks. Panel A illustrates the median ratio of banks' deposits and FHLB advance to their total liabilities over time. Panel B presents the median ratio of the combined total from deposits and FHLB advance to total liabilities.

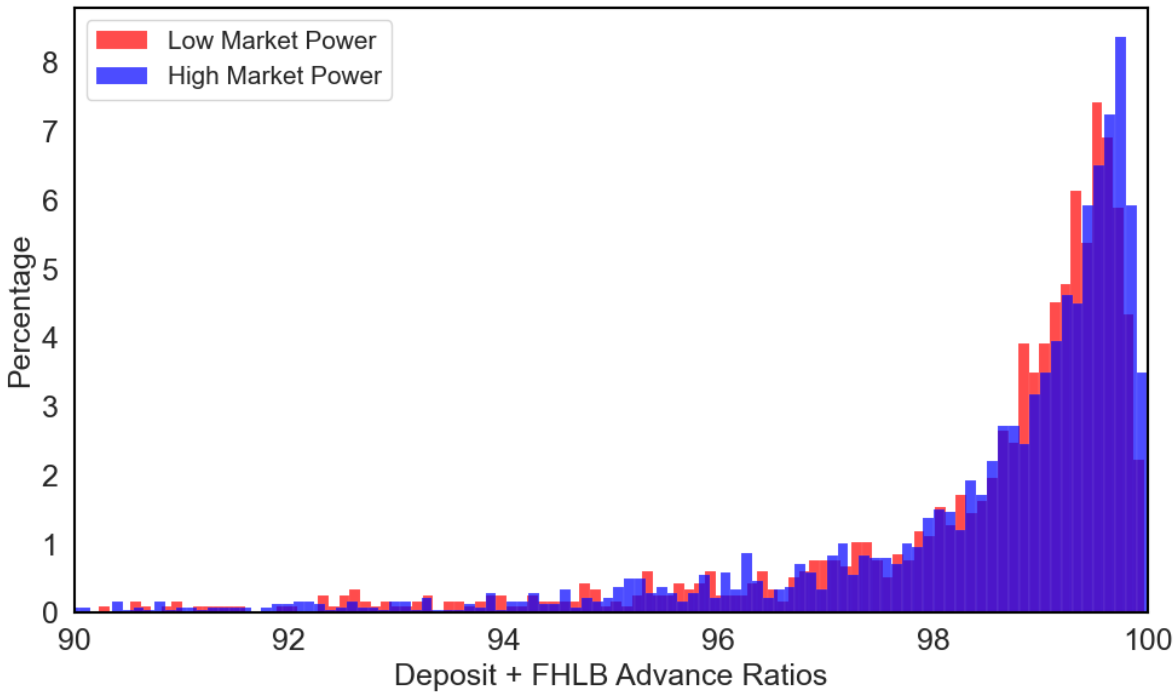

Figure 11: Histogram of Ratio of Deposits and FHLB Advance Combined to Bank Liabilities. In this figure, I plot the histogram of the ratio of deposits and FHLB advance combined to bank liabilities. The y axis is the percentage of banks within the high (low) market power group.

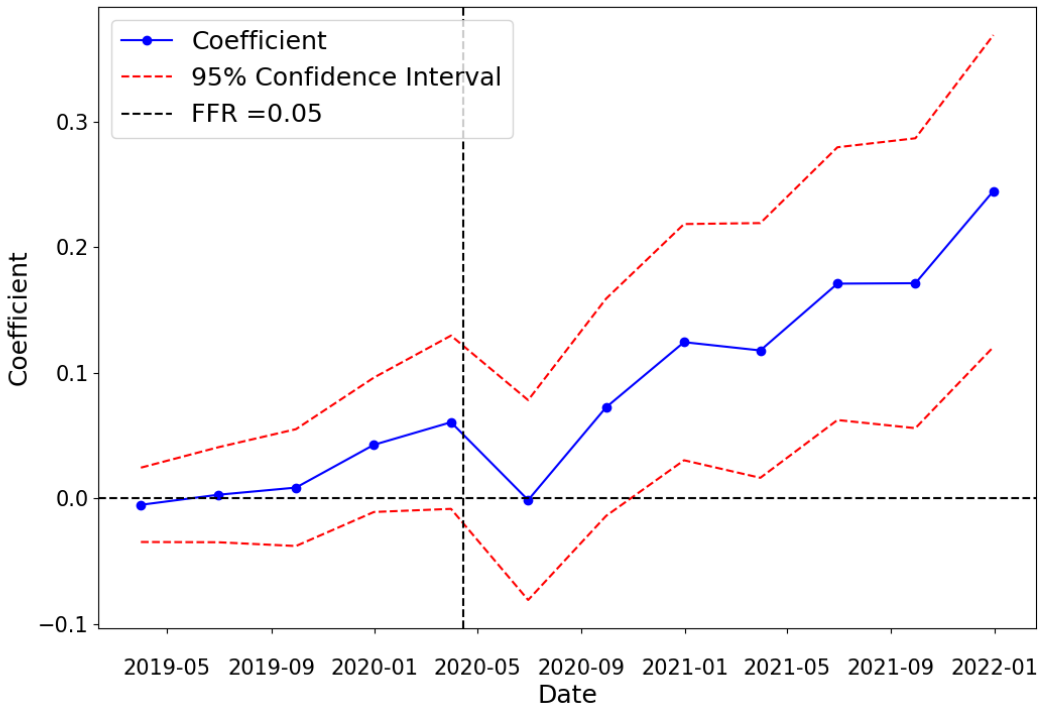

Figure 12: Diff-in-Diff Analysis on Asset Maturities. This graph illustrates the estimates of β_t in the following regression.

$$q_{it} = \sum_{t=2019Q1}^{2021Q4} \beta_t(\text{High MP}_i \times \lambda_t) + \beta_2 \text{High MP}_i + \beta \text{Control}_i + \delta_t + \varepsilon_{it} \tag{40}$$

Here, λ_t equals 1 if date = t and zero otherwise. Coefficients β_t represent difference in asset maturity between high and low deposit market power banks for each quarter relative to 2018Q4. Control_i includes banks' Log asset value, Tier 1 capital ratio, deposit growth, and deposit ratio in 2019Q4.

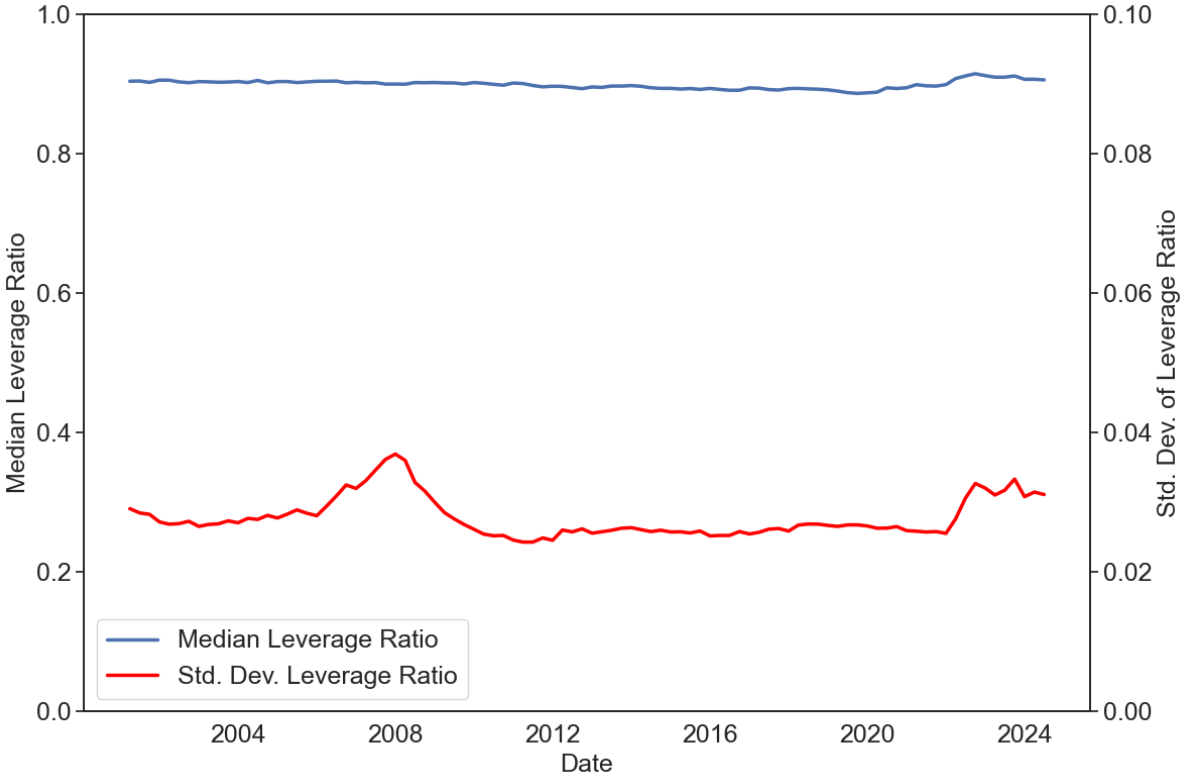

Figure 13: Median and Standard Deviation of Banks' Leverage Ratio. This graph illustrates the median level of banks' leverage ratios alongside the Fed funds rate. The leverage ratio of a bank is defined as the ratio of its liabilities to its assets.

A Appendix for Theoretical Analysis

A.1 A Microfoundation of the Loans Demand Curve

To differentiate the bank discussed in the main text from other lenders, I will refer to it as the "local bank". In the market where the local bank operates, there is a measure a of heterogeneous penniless entrepreneurs, indexed by j. Entrepreneurs have investment projects that require external financing, which they can obtain either from the local bank or from remote lenders, including other banks or non-bank financial institutions. The local bank sets a loan rate to compete with these remote lenders, who are assumed to have no market power in the local market. The opportunity cost of funding for remote lenders is assumed to be r_1 at $t = 1$.

Each entrepreneur has a project that requires a unit investment at t and yields a stochastic return $\tilde{R}$ at $t = 1$ given by

$$\tilde{R} = \begin{cases} R & \text{with probability } 1 - p + mp \\ 0 & \text{with probability } p - mp \end{cases}$$

where $p \in (0,1)$ is a constant parameter, and $m \in \{0,1\}$ is an indicator variable representing the lender's monitoring decision. Monitoring ensures that the project remains safe but incurs a cost. For the bank, the monitoring cost is normalized to zero. However, the relative monitoring cost y_i for remote lenders depends on the opacity of the project—more opaque projects lead to higher monitoring costs for these lenders. To characterize this difference, the relative monitoring cost y_j is assumed to follow a uniform distribution over $[0, \varepsilon]$. For simplicity, it is assumed that the following condition holds.

$$\frac{p}{1-p} \geq \varepsilon \tag{41}$$

$$R - r_1 - \varepsilon - 1 \geq 0 \tag{42}$$

As it will be clear in the later discussion, Equation 41 implies that remote lenders will monitor

as long as their participation constraint is satisfied, Equation 42 implies that all entrepreneur will get finance, either from local bank or remote lenders.

Loans Demand. To characterize the loans demand, I start by considering the case of an entrepreneur of type y_j seeking to borrow from remote lenders. The optimal contract between the remote lenders and the entrepreneur at date t is represented by the pair (r_A^*, m^*) that solves

$$\max_{r_A, m} (1 - p + pm)(R - r_A - 1)$$

subject to the remote lender's incentive compatibility constraint,

$$m^* = argmax_{m \in \{0,1\}} \left[(1 - p + pm)(1 + r_A^*) - my_i\right]$$

the remote lender's participation constraint,

$$(1 - p + pm^*)(1 + r_A^*) - m^* y_i \geq 1 + r_1$$

and the entrepreneur's participation constraint.

$$(1 - p + pm^*)(R - r_A^* - 1) > 0$$

It is clear that the remote lender will lend and monitor if

$$p(1 + r_A^*) > y_i$$

and the participation constraint is satisfied if

$$1 + r_A^* - y_i \geq 1 + r_1$$

Due to the assumption made in Equation 41, we have $\frac{p}{1-p} \geq \varepsilon > y_i$, which implies that,[32]

$$y_i < p(1 + r_1 + y_i)$$

Therefore, as long as the participation constraint of the remote lender is satisfied, the remote lender will monitor the project. Given that the market for remote lenders is competitive, the lending rate for remote lenders is $r_A^* = y_i + r_1$, which is also the outcome of the maximization problem.

The entrepreneur's participation constraint will be satisfied if

$$R(r_1) - r_1 - y_i - 1 > 0$$

which always holds given assumption made in Equation 42. As a result, the market equilibrium rate of borrowing from the remote lender for entrepreneur with type y_i is

$$r_A^* = r_1 + y_i$$

As the local bank has no monitoring cost, it will always monitor the project. The entrepreneur will then choose to borrow from the local bank if $r_L < r_A^*$. This implies that the loan demand function of the local bank can be written as.

$$L(r_L) = a \int_{r_L - r_1}^{\varepsilon} \frac{1}{\varepsilon} dy_i = a[1 - \frac{1}{\varepsilon}(r_L - r_1)]$$

Denote $b = \frac{a}{\varepsilon}$, we have

$$L(r_L) = a - b(r_L - r_1)$$

[32] A medium step is $\frac{y_i}{p} - y_i - 1 < 0 < r_1$

A.2 Model Solutions

Final Equity: With the loans demand function, we can then proceed to solve the optimal loans rate and the final equity by solving the following problem.

$$E_2 = \max_{r_L}(r_L - r_1)L + (1+r_1)E_1 + \pi_D^*(r_1)$$

$$s.t. \quad E_2 \geq \gamma L$$

If the capital constraint is binding, the optimal loans rate, the optimal lending volume, and the final equity is given by

$$r_1^L = r_1 + \frac{a + b\gamma - \sqrt{(a-b\gamma)^2 + 4b[(1+r_1)E_1 + \pi_D^*(r_1)]}}{2b}$$

$$L = \frac{a - b\gamma}{2} + \frac{\sqrt{(a-b\gamma)^2 + 4b[(1+r_1)E_1 + \pi_D^*(r_1)]}}{2}$$

$$E_2(E_1, r_1) = \frac{\gamma}{2}[a - b\gamma + \sqrt{(a-b\gamma)^2 + 4b[(1+r_1)E_1 + \pi_D^*(r_1)]}]$$

Optimal Amount of Long-term Asset and Proposition 1. At the initial date, bankers need to choose between short-term and long-term assets.

$$\max_{Q,S_0} \ \mathbf{E}_0[E_2(E_1, r_1)|r_0]$$

$$E_1 = \frac{(1+r_0)(1+\bar{r}_1)}{1+r_1}Q + (1+r_0)S_0$$

$$Q + S_0 = E_0$$

Solving S_0 and plug it in E_1, we have

$$E_1 = \frac{(1+r_0)(\bar{r}_1 - r_1)}{1+r_1}Q + (1+r_0)E_0$$

The F.O.C with respect to Q is then

$$\int [\frac{b\gamma(1+r_0)(\bar{r}_1-r_1)}{\sqrt{(a-b\gamma)^2+4b[(1+r_1)(1+r_0)E_0+(1+r_0)(\bar{r}_1-r_1)Q^*+\pi_D^*(r_1)]}}]dF(r_1|r_0)=0$$

By substituting the distribution assumption and solving for Q^* that satisifies the F.O.C, we obtain the optimal quantity of long-term assets:

$$Q^* = E_0 + \frac{1}{1+r_0}\left[\frac{\pi_D^*(\bar{r}_1+\Delta)-\pi_D^*(\bar{r}_1-\Delta)}{2\Delta}\right]$$

Normalizing the long-term asset with the initial capital gives the expression in Proposition 1.

$$q^* = \frac{Q^*}{E_0} = 1 + \frac{1}{\bar{E}_1}\left[\frac{\pi_D^*(\bar{r}_1+\Delta)-\pi_D^*(\bar{r}_1-\Delta)}{2\Delta}\right] \tag{43}$$

Where $\bar{E}_1 = (1+r_0)E_0$ is the capital at $t=1$ when the interest rate shock is at its expected level.

Notice that Q^* will also minimize the fluctuation in L, i.e.

$$Q^* = \text{argmin}_Q \quad Var(L|r_0)$$

To illustrate this, notice that

$$Var(L) = \frac{1}{4}Var(\sqrt{(a-b\gamma)^2+4b[(1+r_1)(1+r_0)E_0+(1+r_0)(\bar{r}_1-r_1)Q+\pi_D^*(r_1)]})$$

To save notations, denote

$$\Phi(r_1) = (a-b\gamma)^2+4b[(1+r_1)(1+r_0)E_0+(1+r_0)(\bar{r}_1-r_1)Q+\pi_D^*(r_1)]$$

We then have

$$Var(L) = \frac{1}{4}[E(\Phi(r_1))-E(\sqrt{\Phi(r_1)})^2]$$

Notice that $E(\Phi(r_1))$ is a constant. Therefore, to minimize $Var(L)$, we need to maximize $E(\sqrt{\Phi(r_1)})$. Because,

$$\begin{aligned}
E(\Phi(r_1)) =& \frac{1}{3}\sqrt{(a-b\gamma)^2+4b[(1+\bar{r}_1+\Delta)(1+r_0)E_0-(1+r_0)\Delta Q+\pi_D^*(\bar{r}_1+\Delta)]} \\
&+\frac{1}{3}\sqrt{(a-b\gamma)^2+4b[(1+\bar{r}_1)(1+r_0)E_0+\pi_D^*(\bar{r}_1)]} \\
&+\frac{1}{3}\sqrt{(a-b\gamma)^2+4b[(1+\bar{r}_1-\Delta)(1+r_0)E_0+(1+r_0)\Delta Q+\pi_D^*(\bar{r}_1-\Delta)]}
\end{aligned}$$

We need to solve Q such that

$$\begin{aligned}
\frac{\partial E(\Phi(r_1))}{\partial Q} =& \frac{1}{6}\frac{-4b(1+r_0)\Delta}{\sqrt{(a-b\gamma)^2+4b[(1+\bar{r}_1+\Delta)(1+r_0)E_0-(1+r_0)\Delta Q+\pi_D^*(\bar{r}_1+\Delta)]}} \\
&+\frac{1}{6}\frac{4b(1+r_0)\Delta}{\sqrt{(a-b\gamma)^2+4b[(1+\bar{r}_1-\Delta)(1+r_0)E_0+(1+r_0)\Delta Q+\pi_D^*(\bar{r}_1-\Delta)]}} \\
=& 0
\end{aligned}$$

Which gives that

$$Q^* = E_0 + \frac{1}{1+r_0}\left[\frac{\pi_D^*(\bar{r}_1+\Delta)-\pi_D^*(\bar{r}_1-\Delta)}{2\Delta}\right]$$

Therefore, the optimal amount of long-term asset minimizes the fluctuation in lending.

Deposit Market Related Solutions. Given household's second stage utility maximization problem as described in Equation 2, 3, and 4. Assuming the bank don't offer zero deposit rates, it is easy to solve that the optimal consumption and savings are:

$$c_1 = \frac{1}{1+\beta}y \tag{44}$$

$$c_2 = \frac{\beta}{1+\beta}(1+r_D^j)y \tag{45}$$

$$d = \frac{\beta}{1+\beta}y \tag{46}$$

and the indirect utility function conditional on borrowing from bank j is

$$v_j = \ln(\frac{1}{1+\beta}) + \beta \ln(\frac{\beta}{1+\beta}) + (1+\beta)\ln y + \beta \ln(1+r_D^j) \tag{47}$$

With the distributional assumption, tt follows that the probability for a consumer to choose bank j and the deposit demand for bank j is given by:

$$Pr(j) = Pr(V_j = max_r V_r) = \frac{e^{v_j/\mu}}{\int_0^1 e^{v_r/\mu} dr} = \frac{(1+r_D^j)^{\frac{\beta}{\mu}}}{\int_0^1 (1+r_D^r)^{\frac{\beta}{\mu}} dr} = (\frac{1+r_D^j}{1+r_D})^{\frac{\beta}{\mu}} \tag{48}$$

and the deposit demand for a bank is

$$d_j = (\frac{1+r_D^j}{1+r_D})^{\frac{\beta}{\mu}} \frac{\beta}{1+\beta} y \tag{49}$$

Where $1+r_D = (\int_0^1 (1+r_D^r)^{\frac{\beta}{\mu}} dr)^{\frac{\mu}{\beta}}$ represents the aggregate deposit rate.

If the bank offers negative deposit rate, household will hold cash instead of deposits and the deposit demand will be zero.

Proposition 2. The deposit beta can be written as

$$\beta^D = \frac{r_D^*(\bar{r}_1 + \Delta) - r_D^*(\bar{r}_1 - \Delta)}{2\Delta} \tag{50}$$

Plugging the optimal deposit rate function 22, it is easy to show that (1) When $E(r_1|r_0) - \Delta > \frac{\mu}{\beta}$, $\beta^D = \frac{\beta}{\beta+\mu}$, which is constant; (2) When $E(r_1|r_0) + \Delta \leq \frac{\mu}{\beta}$, the deposit beta is zero for all banks; (3) when $E(r_1|r_0) \in \left[\frac{\mu}{\beta} - \Delta, \frac{\mu}{\beta} + \Delta\right]$,

$$\beta^D = \frac{\frac{\beta}{\beta+\mu}(1+\bar{r}_1+\Delta) - 1}{2\Delta} \tag{51}$$

which increases with $\bar{r}_1$. The result follows as $\bar{r}_1$ increases with r_0.

Proposition 3. This proposition immediately follows from Proposition 2.

Proposition 4. Consider two banks with two different deposit market power levels, μ^H and μ^L with $\mu^H > \mu^L$. When $\bar{r}_1 = E(r_1|r_0)$ is sufficiently high, the difference in deposit beta is constant, i.e.

$$\beta^D(\mu^H) - \beta^D(\mu^L) = \frac{\beta}{\beta + \mu^H} - \frac{\beta}{\beta + \mu^L} \tag{52}$$

Now, when $\bar{r}_1$ decreases to a level so that $\frac{\mu^L}{\beta} < \bar{r}_1 - \Delta < \frac{\mu^H}{\beta}$, the difference in deposit beta can be written as

$$\beta^D(\mu^H) - \beta^D(\mu^L) = \frac{\frac{\beta}{\beta+\mu^H}(1 + \bar{r}_1 + \Delta) - 1}{2\Delta} - \frac{\beta}{\beta + \mu^L} \tag{53}$$

It follows that, within this future risk-free rate region, as $\bar{r}_1$ decreases, the deposit beta declines more for the bank with greater deposit market power. It then follows that the asset maturities increase more for the bank with greater deposit market power.

A.3 Alternative Probability Distribution

Given the empirical persistence of the policy rate and the low probability of having negative policy rate in US, the change in risk-free rate (i.e. Δ) may not be symmetry. In this subsection, I show that the main result still holds. In particular, the F.O.C for the long-term asset holding is

$$\int [\frac{b\gamma(1 + r_0)(\bar{r}_1 - r_1)}{\sqrt{(a - b\gamma)^2 + 4b[(1 + r_1)(1 + r_0)E_0 + (1 + r_0)(\bar{r}_1 - r_1)Q^* + \pi_D^*(r_1)]}}]dF(r_1|r_0) = 0$$

Instead of the symmetric policy rate shock, I assume that the policy rate shock is asymmetric. Specifically, it is assumed that

- With probability $\frac{1}{mt}$, $r_1 = \bar{r}_1 + t\Delta$.
- With probability $1 - \frac{1}{mt} - \frac{1}{m}$, $r_1 = \bar{r}_1$.
- With probability $\frac{1}{m}$, $r_1 = \bar{r}_1 - \Delta$.

Plug in the distribution assumption and solve for the F.O.C., we have the optimal amount of long-term asset as

$$
\begin{aligned}
Q^* &= E_0 + \frac{1}{1+r_0}\left[\frac{\pi_D^*(\bar{r}_1 + t\Delta) - \pi_D^*(\bar{r}_1 - \Delta)}{(t+1)\Delta}\right] \\
&\approx E_0 + \frac{1}{1+r_0}\left[\frac{\partial \pi_D^*}{\partial r_1}\Big|_{r_1=\bar{r}_1}\right]
\end{aligned}
$$

where the approximation becomes exact as $\Delta \to 0$. Notice that this delivers the same economics intuition as in the symmetric case, where

$$
\begin{aligned}
Q^* &= E_0 + \frac{1}{1+r_0}\left[\frac{\pi_D^*(\bar{r}_1 + \Delta) - \pi_D^*(\bar{r}_1 - \Delta)}{2\Delta}\right] \\
&\approx E_0 + \frac{1}{1+r_0}\left[\frac{\partial \pi_D^*}{\partial r_1}\Big|_{r_1=\bar{r}_1}\right]
\end{aligned}
$$

B Appendix for Empirical Analysis

B.1 Details of Maturity Variables and Their Coverage Over Time

In this section, I summarize the relevant maturity variables and their corresponding Call Report codes in Tables B.1 to B.5. The Call Report presents maturity information in intervals; for instance, the maturity of a security might be described as 'Over three months through 12 months.' In the summary tables, I delineate these intervals for each asset or liability, include the associated Call Report code, and specify the assigned maturities. All assigned maturity numbers are standard values commonly used in recent literature (e.g Drechsler, Savov, and Schnabl (2021) and English, Van den Heuvel, and Zakrajšek (2018)).

This maturity information covers the majority of banks' assets and liabilities. As shown in Figure B.1, the median level of maturity information coverage was more than 95% for assets and more than 99% for liabilities during my sample period for the diff-in-diffs analysis.

B.2 Bank Level ROA Betas

In this section, I estimate ROA beta at the bank level for both high- and low-interest-rate environment. Specifically, I estimate the ROA beta for each bank i in both (relatively) high and low interest rate environment using the following regression:

$$\Delta\text{ROA}_{i,t} = \alpha_i + \eta_t + \sum_{\tau=0}^{3} \beta_{i,\tau}^{ROA} \Delta r_{t-\tau} + \varepsilon_{it} \tag{54}$$

where $\Delta\text{ROA}_{i,t}$ is the change in bank i's ROA at date t, Δr_t is the change in the Fed funds rate at date t, and α_i and η_t denote the bank and time fixed effects, respectively. The ROA beta of bank i is the sum of the coefficients $\beta_i^{ROA} = \sum_{\tau=0}^{3} \beta_{i,\tau}^{ROA}$. Given that low-interest-rate environments lasted for 13 years, from 2009 to 2021, I used data from this period to estimate the ROA betas for low-interest-rate environments. To ensure a meaningful comparison, I estimated the ROA betas for the (relatively) high interest rate environment using data from the preceding 13 years, specifically from 1996 to 2008. As shown in Table B.6, the average estimated ROA beta is close to zero for both high and low interest rate environments. Additionally, the estimated beta can be either greater or smaller than zero, which reflects banks' efforts to hedge the interest rate risk of their earnings.

B.3 Estimation Details of the Mixture Model

Remember that the underlying data generating process is assumed to follow a mixture distribution, where the observation of bank i's deposit rates $\boldsymbol{r}_i^d$ is drawn from the following process with probability π_k,

$$\boldsymbol{r}_i^d = \xi_i \boldsymbol{e} + \beta_k \boldsymbol{r} + \boldsymbol{\varepsilon}_i$$

where $\boldsymbol{r}_i^d$, $\boldsymbol{e}$, and $\boldsymbol{r}$ are $T \times 1$ vectors, $\boldsymbol{e} = [1, 1, ..., 1]'$, $\boldsymbol{r}_i^d = [r_{i,1}^d, r_{i,2}^d, ..., r_{i,T}^d]'$, $\boldsymbol{r} = [r_1, r_2, ..., r_T]'$, and $\boldsymbol{\varepsilon}_i \sim \mathcal{N}(0, \sigma^2 \boldsymbol{I})$.

For each k, notice that we can transform the data by subtracting time series average and get

$$r_{i,t}^d - \frac{1}{T}\sum_{t=1}^{T} r_{i,t}^d = \beta_k(r_t - \frac{1}{T}\sum_{t=1}^{T} r_t) + (\varepsilon_{i,t} - \frac{1}{T}\sum_{t=1}^{T} \varepsilon_{i,t})$$

To facilitate analysis, I denote

$$\tilde{\boldsymbol{r}}_i^d = \boldsymbol{r}_i^d - (\frac{1}{T}\sum_{t=1}^{T} r_{i,t}^d)\boldsymbol{e}$$

$$\tilde{\boldsymbol{x}} = \boldsymbol{r} - (\frac{1}{T}\sum_{t=1}^{T} r_t)\boldsymbol{e}$$

$$\tilde{\boldsymbol{\varepsilon}}_i = \boldsymbol{\varepsilon}_i - (\frac{1}{T}\sum_{t=1}^{T} \varepsilon_{i,t})\boldsymbol{e}$$

$$\tilde{\boldsymbol{r}}_i^d = \tilde{\boldsymbol{x}}\beta_k + \tilde{\boldsymbol{\varepsilon}}_i$$

$$\tilde{\sigma}^2 = \frac{T-1}{T}\sigma^2$$

Therefore, we can work on the revised problem where the observation $\tilde{\boldsymbol{r}}_i^d$ is drawn from the following data generating process with probability π_k,

$$\tilde{\boldsymbol{r}}_i^d = \tilde{\boldsymbol{x}}\beta_k + \tilde{\boldsymbol{\varepsilon}}_i$$

where $\sum_k \pi_k = 1$, and $\tilde{\boldsymbol{\varepsilon}}_i \sim \mathcal{N}(0, \tilde{\sigma}^2\boldsymbol{I})$.

With these information, we can apply the EM algorithm by using the following steps.

1. Pick the initial adaptive parameters (θ), i.e. σ^2 and for each k, π_k, β_k.

2. Calculate $\tilde{\boldsymbol{r}}_i^d$, $\tilde{\boldsymbol{x}}$, and $\tilde{\sigma}^2$.

3. (E Step:) Calculate the posterior distribution

$$\gamma_{ik} = \frac{\pi_k \mathcal{N}(\tilde{\boldsymbol{r}}_i^d | \tilde{\boldsymbol{x}}\beta_k, \tilde{\sigma}^2\boldsymbol{I})}{\sum_j \pi_j \mathcal{N}(\tilde{\boldsymbol{r}}_i^d | \tilde{\boldsymbol{x}}\beta_j, \tilde{\sigma}^2\boldsymbol{I})} = \frac{\pi_k exp(-\frac{1}{2\tilde{\sigma}^2}(\tilde{\boldsymbol{r}}_i^d - \tilde{\boldsymbol{x}}\beta_k)'(\tilde{\boldsymbol{r}}_i^d - \tilde{\boldsymbol{x}}\beta_k))}{\sum_{j=1}^{K} \pi_j exp(-\frac{1}{2\tilde{\sigma}^2}(\tilde{\boldsymbol{r}}_i^d - \tilde{\boldsymbol{x}}\beta_j)'(\tilde{\boldsymbol{r}}_i^d - \tilde{\boldsymbol{x}}\beta_j))}$$

4. (M Step:) Maximize the following function with respect to the adaptive parameters:

$$Q(\theta,\theta^{old}) = \sum_{i=1}^{N}\sum_{k=1}^{K}\gamma_{ik}[\ln\pi_k + \ln\mathcal{N}(\tilde{\boldsymbol{r}}_i^d|\tilde{\boldsymbol{x}}\beta_k,\tilde{\sigma}^2\boldsymbol{I})]$$

with respect to θ, keep the γ_{ik} fixed. Because π_k is from a probability density function, we know that $\sum\pi_k = 1$. With this constraint, it can be shown that

$$\hat{\pi}_k = \frac{1}{N}\sum_{i=1}^{N}\gamma_{ik}$$

Consider first the maximization with respect to the parameter β_k of the kth model, it can be shown that the function $Q(\theta,\theta^{old})$ takes the following form

$$Q(\theta,\theta^{old}) = \sum_{i=1}^{N}\gamma_{ik}(-\frac{1}{2\tilde{\sigma}^2}(\tilde{\boldsymbol{r}}_i^d - \tilde{\boldsymbol{x}}\beta_k)'(\tilde{\boldsymbol{r}}_i^d - \tilde{\boldsymbol{x}}\beta_k)) + \text{const}$$

Setting the derivative with respect to β_k equal to zero gives

$$\hat{\beta}_k = [\sum_{i=1}^{N}\gamma_{ik}\tilde{\boldsymbol{x}}'\tilde{\boldsymbol{x}}]^{-1}[\sum_{i=1}^{N}\gamma_{ik}\tilde{\boldsymbol{x}}'\tilde{\boldsymbol{r}}_i^d]$$

Finally, I maximize $Q(\theta,\theta^{old})$ with respect to $\tilde{\sigma}^2$. Keeping only terms that depend on $\tilde{\sigma}^2$, the function $Q(\theta,\theta^{old})$ can be written as

$$Q(\theta,\theta^{old}) = \sum_{i=1}^{N}\sum_{k=1}^{K}\gamma_{ik}[-\frac{T}{2}ln(\tilde{\sigma}^2) - \frac{1}{2\tilde{\sigma}^2}(\tilde{\boldsymbol{r}}_i^d - \tilde{\boldsymbol{x}}\beta_k)'(\tilde{\boldsymbol{r}}_i^d - \tilde{\boldsymbol{x}}\beta_k)]$$

Setting the derivative with respect to $\tilde{\sigma}^2$ equals to zero, and use the estimates for β_k, the estimates of σ^2 can be written as

$$\hat{\tilde{\sigma}}^2 = \frac{1}{NT}\sum_{i=1}^{N}\sum_{k=1}^{K}\gamma_{ik}(\tilde{\boldsymbol{r}}_i^d - \tilde{\boldsymbol{x}}\hat{\beta}_k)'(\tilde{\boldsymbol{r}}_i^d - \tilde{\boldsymbol{x}}\hat{\beta}_k)$$

5. Using the updated parameters, I can check for convergence. If the likelihood function

does not converge, I assign $\theta^{old} = \theta$ and go back to step 3.

B.4 Main Results Using Classification Based on Bank-Level Estimations

In this section, I conduct a robustness check using group classification based on bank-level estimations. First, I estimate Equation 30 using data from the 2015–2018 monetary policy tightening period. Banks with estimated β_i values greater than the median are classified as low deposit market power banks, while banks with estimated β_i values below the median are classified as high deposit market power banks. I then repeat the analysis of the expense beta and asset maturity using these classification results.

The results for both the expense beta and asset maturity are consistent with the main analysis presented in Section 4. As shown in Table B.7, the expense beta decreases for both groups of banks when the economy transitions to a low-interest-rate environment, with a significantly larger decrease observed for banks with higher deposit market power. Similarly, Table B.8 shows that asset maturity increased for both groups of banks after 2020Q2, with a significantly larger increase for banks with higher deposit market power. The magnitude of the differential changes in both the expense beta and asset maturity are weaker than the results presented in the main analysis, consistent with the idea that classifications based on median bank-level estimations are less precise.

B.5 Maturity Gap

In the main analysis, I use banks' asset maturity as the primary variable of interest. This choice is guided by my theoretical model and the understanding that banks' asset choices are influenced by the combined cash flows from multiple deposit product business lines. In the literature (e.g., Drechsler, Savov, and Schnabl (2021) and English, Van den Heuvel, and Zakrajšek (2018)),

another commonly used measure of banks' interest rate risk is the maturity gap, i.e.

$$g_{it} = \sum_{j=1}^{N_1} \omega_{A,it}^{j} rm_{A,it}^{j} - \sum_{k=1}^{N_2} \omega_{L,it}^{k} rm_{L,it}^{k} \tag{55}$$

where g_{it} denotes the repricing maturity gap for bank i at date t, $\omega_{A,it}^{j}$ is the portfolio weight for asset class j and $rm_{A,it}^{j}$ is the corresponding average repricing maturity. Similarly, $\omega_{L,it}^{k}$ is the portfolio weight for liability class k and $rm_{L,it}^{k}$ is the corresponding average repricing maturity. Intuitively, the maturity gap captures the difference in interest rate sensitivity between banks' assets value and liabilities value, thereby reflecting interest rate risk from balance sheet but excluding those associated with deposit market power. Therefore, banks can adjust their maturity gap to offset changes in risk associated with their deposit market power.

In this section, I demonstrate that all my results remain consistent when using maturity gap as the variable of interest. In figure B.2, I plotted the time series of the banking industry's median maturity gap, alongside with Fed funds rate. As the figure shows, the pattern of the median maturity gap in the banking industry is very similar to the pattern of asset maturity, i.e. it increased when the Fed funds rate dropped to zero both after the global financial crisis and the Covid-19 Pandemic.

I then proceed to conduct the diff-in-diffs analysis by running the following regressions:

$$g_{it} = \beta_1(\text{High MP}_i \times \text{Post}_t) + \beta_2 \text{High MP}_i + \beta_3 \text{Post}_t + \text{Control}_i + \varepsilon_{it} \tag{56}$$

$$g_{it} = \beta_1(\text{High MP}_i \times \text{Post}_t) + \alpha_i + \delta_t + \varepsilon_{it} \tag{57}$$

As Table B.9 shows, the maturity gap increased for both groups of banks after the Fed funds rate dropped to zero. However, high deposit market power banks experienced a significantly larger increase in maturity gap compared to low deposit market power banks. Specifically, following the ultra-low Fed funds rate after 2020Q2, the maturity gap of low deposit market power banks increased by 0.416 years (5.00 months), while the maturity gap for high deposit market power

banks increased by an additional 0.134 years (1.61 months). The regression also confirms that the maturity gap is higher for high deposit market power banks.

Maturity Interval	Call Report Code	Maturity Used
Panel A: U.S. Government Securities and other non-mortgage debt securities; and MBS other than those in 1.b		
(0,0.25)	RCFDA549	0.125
[0.25,1)	RCFDA550	0.625
[1,3)	RCFDA551	2
[3,5)	RCFDA552	4
[3,15)	RCFDA553	10
[15,)	RCFDA554	20
Panel B: MBS backed by closed-end first lien 1-4 family residential mortgages		
(0,0.25)	RCFDA555	0.125
[0.25,1)	RCFDA556	0.625
[1,3)	RCFDA557	2
[3,5)	RCFDA558	4
[3,15)	RCFDA559	10
[15,)	RCFDA560	20
Panel C: Other mortgage-backed securities		
(0,3)	RCFDA561	1.5
[3,)	RCFDA562	5

Table B.1: Maturities Information for Securities.

Maturity Interval	Call Report Code	Maturity Used
Panel A: Mortgage Secured by first liens on 1-4 family residential properties		
(0,0.25)	RCONA564	0.125
[0.25,1)	RCONA565	0.625
[1,3)	RCONA566	2
[3,5)	RCONA567	4
[3,15)	RCONA568	10
[15,)	RCONA569	20
Panel B: All loans and leases excluding Panel A		
(0,0.25)	RCFDA570	0.125
[0.25,1)	RCFDA571	0.625
[1,3)	RCFDA572	2
[3,5)	RCFDA573	4
[3,15)	RCFDA574	10

Table B.2: Maturities Information for Loans.

Asset Name	**Call Report Code**	**Maturity**
Noninterest-bearing Cash	RCFD0081	0
Interest-bearing Cash	RCFD0071	0
Fed Funds Sold	RCONB987	0
Repo Purchased	RCFDB989	0

Table B.3: Zero Maturity Assets

Maturity Interval	Call Report Code	Maturity
Panel A: Maturity and repricing data for time deposits		
(0,0.25)	RCONHK07+RCONHK12	0.125
[0.25,1)	RCONHK08+RCONHK13	0.625
[1,3)	RCONHK09+RCONHK14	2
[3,)	RCONHK10+RCONHK15	5
Panel B: Federal Home Loan Bank advances		
(0,1)	RCFDF055	0.5
[1,3)	RCFDF056	2
[3,5)	RCFDF057	4
[5,)	RCFDF058	5
Panel C: Other borrowings		
(0,1)	RCFDF060	0.5
[1,3)	RCFDF061	2
[3,5)	RCFDF062	4
[5,)	RCFDF063	5
Panel D: Subordinated notes and debentures		
Asset	**Call Report Code**	**Maturity**
Subordinated notes and debentures	RCFD3200	5

Table B.4: Maturity Information for Positive Maturity Liabilities

Table B.6: Summary Statistics of the Estimated ROA Betas.

Sample Period	Mean	Std	25%	50%	75%	Obs
1996 to 2008	0.07	0.31	-0.12	0.01	0.19	4,171
2009 to 2021	-0.05	0.67	-0.34	-0.03	0.29	4,086

Liability Name	**Call Report Code**	**Maturity**
Transaction Deposit	RCON2215	0
Savings deposits (MMDA)	RCON6810	0
Other Saving	RCON0352	0
Fed Fund Purchased	RCONB993	0
Repo Sold	RCFDB995	0

Table B.5: Zero Maturity Liabilities

Table B.7: Deposit Beta in (Relatively) High and Low Interest Rate Environments for High and Low Deposit Market Power Banks (Classification Using Bank-Level Estimations). This table summarizes the results from running the following regression:

$$\Delta r_{i,t}^{COF} = \alpha_i + \sum_{\tau=0}^{3} \xi_\tau^D \Delta r_{t-\tau} \mathbb{1}(r_{t-\tau} \leq \tilde{r}) \text{HighMP}_i + \sum_{\tau=0}^{3} \gamma_\tau^D \Delta r_{t-\tau} \mathbb{1}(r_{t-\tau} \leq \tilde{r})$$
$$+ \sum_{\tau=0}^{3} \theta_\tau^D \Delta r_{t-\tau} \text{HighMP}_i + \sum_{\tau=0}^{3} \beta_\tau^D \Delta r_{t-\tau} + \beta \text{Control}_{i,t} + \varepsilon_{i,t}$$

$\beta^D = \sum_{\tau=0}^{3} \beta_\tau^D$ represents the deposit beta for banks with low deposit market power when the risk-free rates are relatively high. $\theta^D = \sum_{\tau=0}^{3} \theta_\tau^D$ signifies the difference in deposit beta between high deposit market power banks and low deposit market power banks when the risk-free rate is relatively high. $\gamma^D = \sum_{\tau=0}^{3} \gamma_\tau^D$ indicates the change in deposit beta when comparing low interest rate environments to period when the risk-free rates are relatively high for low deposit market power banks. $\xi^D = \sum_{\tau=0}^{3} \xi_\tau^D$ denotes the additional change in deposit beta for high deposit market power banks compared to low deposit market power banks when comparing low interest rate environments to period when the risk-free rates are relatively high. $\mathbb{1}(r_{t-\tau} \leq \tilde{r})$ equals 1 if r_t is less than $\tilde{r}$, and 0 otherwise. The sample period is 2014Q2-2024Q2. The classification of the deposit market power group is based on the bank-level estimation of Equation 30, with details provided in Section B.4. The threshold $\tilde{r}$ is 1.5%. Standard errors are double-clustered at the bank and time levels and are reported in parentheses. * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

	(1)	(2)	(3)
$\tilde{r}$	1.5%	1%	2%
β^D	0.432***	0.436***	0.431***
	(0.015)	(0.014)	(0.017)
γ^D	−0.127***	−0.184***	−0.098***
	(0.035)	(0.036)	(0.037)
θ^D	−0.120***	−0.125***	−0.118***
	(0.011)	(0.010)	(0.012)
ξ^D	−0.055***	−0.046*	−0.047**
	(0.021)	(0.024)	(0.019)
Control	Yes	Yes	Yes
Bank FE	Yes	Yes	Yes
R^2	0.5311	0.5355	0.5300
Num of Obs	139,966	139,966	139,966

Table B.8: Diff-in-Diff Analysis on Asset Maturity (Classification Using Bank-Level Estimations). This table summarizes the results from running the following regressions:

$$q_{it} = \beta_1(\text{High MP}_i \times \text{Post}_t) + \beta_2 \text{High MP}_i + \beta_3 \text{Post}_t + \beta \text{Control}_i + \varepsilon_{it}$$

$$q_{it} = \beta_1(\text{High MP}_i \times \text{Post}_t) + \alpha_i + \delta_t + \varepsilon_{it}$$

q_{it} is bank i's asset maturity at date t. High MP_i is a binary variable that indicates whether bank i belongs to the high deposit market power group. Post_t is a binary indicator that denotes whether the date t is post 2020Q2. The classification of the deposit market power group is based on the bank-level estimation of Equation 30, with details provided in Section B.4. The sample period spans from 2018Q4 to 2021Q4. The standard errors are clustered at the bank level and are included in the parentheses. * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

	(1)	(2)
HighMP × Post	0.086**	0.070**
	(0.037)	(0.034)
Post	0.455***	
	(0.025)	
HighMP	0.574***	
	(0.083)	
Controls	Yes	No
Bank FE	No	Yes
Time FE	No	Yes
R^2	0.1001	0.9294
Num of Obs	43,611	44,393

Table B.9: Diff-in-Diff Analysis on Maturity Mismatch. This table summarizes the results from running the following regressions:

$$g_{it} = \beta_1(\text{High MP}_i \times \text{Post}_t) + \beta_2\text{High MP}_i + \beta_3\text{Post}_t + \text{Control}_i + \varepsilon_{it}$$

$$g_{it} = \beta_1(\text{High MP}_i \times \text{Post}_t) + \alpha_i + \delta_t + \varepsilon_{it}$$

g_{it} is bank i's maturity gap at date t. High MP_i is a binary variable that indicates whether bank i belongs to the high deposit market power group. Post_t is a binary indicator that denotes whether the date t is post 2020Q2. The classification of the deposit market power group is based on the mixture method. The sample period is from 2018Q4 to 2021Q4. The standard errors are clustered at the bank level and are included in the parentheses. $^{*}\,p < 0.1$; $^{**}\,p < 0.05$; $^{***}\,p < 0.01$.

	(1)	(2)	(4)
HighMP × Post	0.139***	0.134***	0.090**
	(0.041)	(0.039)	(0.036)
Post	0.516***	0.416***	
	(0.033)	(0.031)	
HighMP	0.839***	0.801***	
	(0.089)	(0.092)	
Controls	No	Yes	No
Time FE	No	No	Yes
Bank FE	No	No	Yes
R^2	0.0421	0.1048	0.9302
Observations	40,723	39,932	40,671

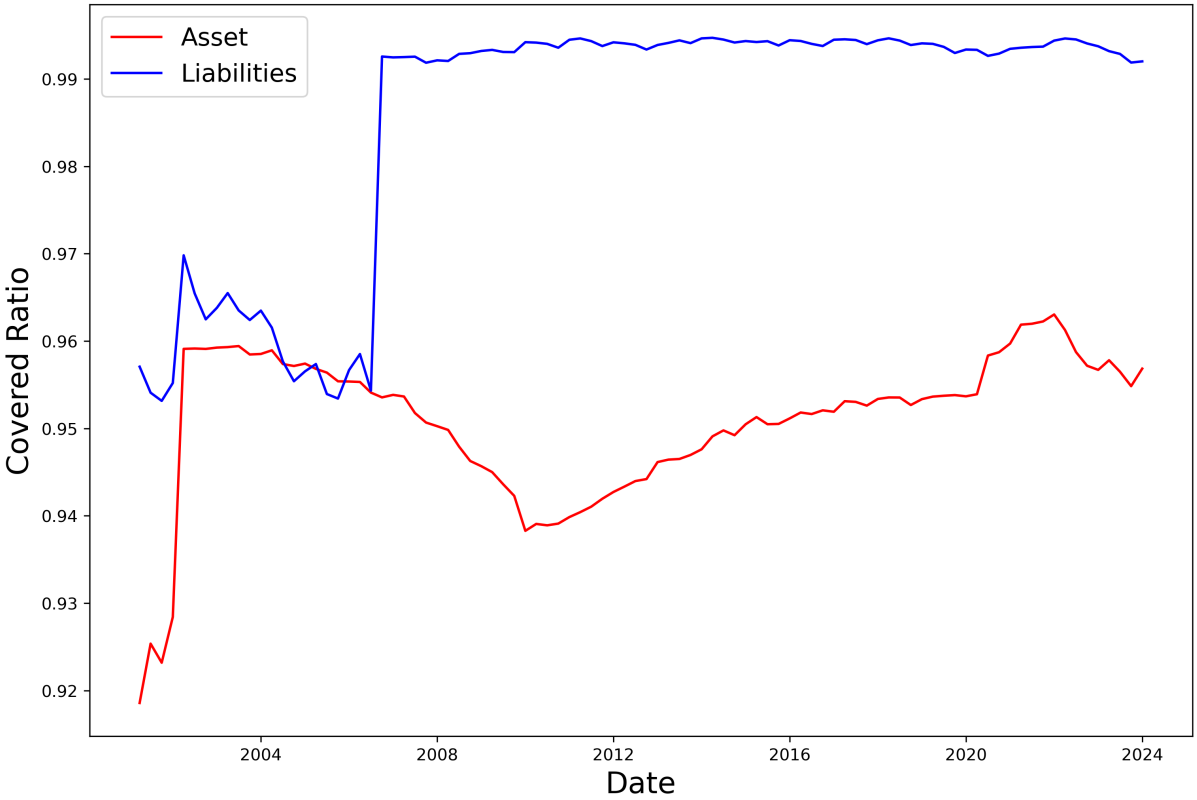

Figure B.1: The Median Ratio of Assets and Liabilities Got Covered. This figure illustrates the median level of maturity information coverage for assets and liabilities as reported in the Call Report.

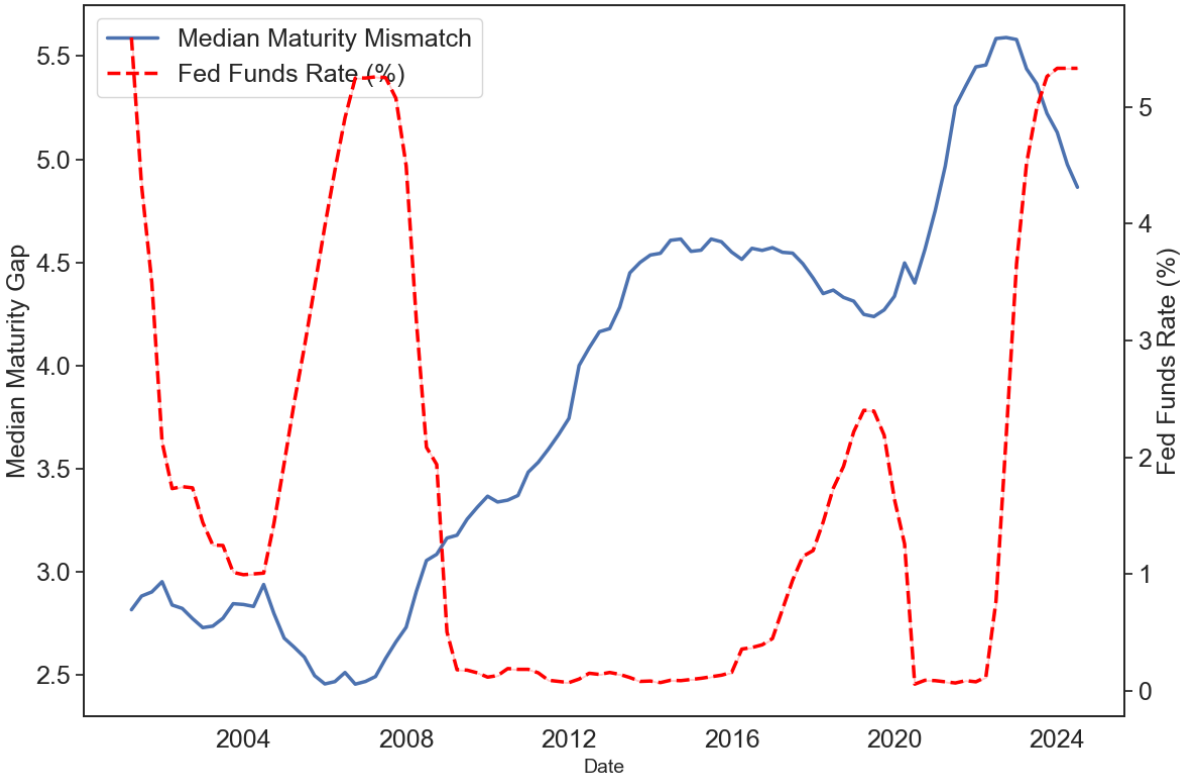

Figure B.2: Median Maturity Gap. This graph illustrates the median level of banks' maturity gap, alongside the Federal Funds rate. The maturity gap measures the difference in the sensitivity of banks' asset and liabilities, based on the balance sheet variables.